Online Appendix

A Model of the Attention Economy: Data, Platforms, and Discovery

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B Supplemental Material for Section 4

Appendix B.1 gives the formal statement of a firm's bidding problem. Appendix B.2 proves the auxiliary lemmas used in the proof of Theorem 1.

B.1 Formal Statement of Firms' Bidding Problem

Let τ_z denote the zth time of entry into an auction for consumer i. That is, τ_z is the zth arrival of a Poisson process that ticks at rate $\lambda_{at} + \lambda_f \mathbb{1}_{\{j \in \Omega_{it}\}}$. Taking the bidding strategy $B_t(\cdot)$ of its rivals as given, firm j sets bids to maximize the present value of flow profits (including the costs of advertising) from selling to a typical consumer i:

$$\Pi_{\mathbb{J}} = \max_{\{b_z\}} \mathbb{E}\left[\int_0^{\infty} e^{-\rho s} \pi_{\mathbb{J}s}(\hat{v}_{ij}) \mathbb{1}_{\{j \in \Omega_{is}\}} \, \mathrm{d}s - \sum_{z=1}^{\infty} e^{-\rho \tau_z} B_{\tau_z}\left(\hat{v}_z^{(1)}\right) \mathbb{1}_{\left\{b_z > B_{\tau_z}\left(\hat{v}_z^{(1)}\right)\right\}}\right]$$

where $\hat{v}_z^{(1)}$ is the highest expectation of the $N-1$ other bidders in the zth auction: $\hat{v}_z^{(1)} \sim \left(H_{\tau_z}^c\right)^{N-1}$ conditional on τ_z. Above, the bid b_z in the zth auction is a measurable function of the expectation $\hat{v}_{ij}$ and time t.

B.2 Auxiliary Lemmas for Theorem 1

This subsection proves Lemmas 2–5, used in the proof of Theorem 1.

Lemma 1. *In an equilibrium, the bidding function $B_t(\cdot)$ is Lipschitz at each t.*

Proof. Since $B_t = V_t^{\text{In}} - V_t^{\text{Out}}$, it suffices to show that V_t^{In} and V_t^{Out} are Lipschitz. I argue for V_t^{In}; the argument for V_t^{Out} is analogous.

Fix two expected values $\hat{v} > \tilde{v}$. Consider a firm with expected value $\tilde{v}$ that deviates by bidding as if its expected value were $\hat{v}$. The deviating firm's bidding induces the same dynamics for $\mathbb{1}_{\{j \in \Omega_{is}\}}$ as a firm actually with expected value $\hat{v}$, but its true flow profit at time s is $\pi_{\mathbb{J}s}\tilde{v}$ rather than $\pi_{\mathbb{J}s}\hat{v}$. Letting $\hat{V}_t^{\text{In}}(\tilde{v})$ denote the value of this deviation,

$$V_t^{\text{In}}(\hat{v}) - \hat{V}_t^{\text{In}}(\tilde{v}) = (\hat{v} - \tilde{v})\, \mathbb{E}\left[\int_t^\infty e^{-\rho(s-t)} \pi_{\mathbb{J}s}\, \mathbb{1}_{\{j \in \Omega_{is}\}}\, \mathrm{d}s\right],$$

where the expectation is taken under the $\hat{v}$-bidding law. Note that ad costs cancel since both firms bid identically.

Optimality of $V_t^{\text{In}}(\tilde{v})$ implies $V_t^{\text{In}}(\tilde{v}) \geq \hat{V}_t^{\text{In}}(\tilde{v})$, so

$$V_t^{\text{In}}(\hat{v}) - V_t^{\text{In}}(\tilde{v}) \leq (\hat{v} - \tilde{v})\, \mathbb{E}\left[\int_t^\infty e^{-\rho(s-t)} \pi_{\mathbb{J}s}\, \mathbb{1}_{\{j \in \Omega_{is}\}}\, \mathrm{d}s\right] \leq \frac{\sup_{s \geq t} \pi_{\mathbb{J}s}}{\rho}\, (\hat{v} - \tilde{v}).$$

Since $V_t^{\text{In}}(\hat{v}) \geq V_t^{\text{In}}(\tilde{v})$ by monotonicity (a firm with higher expected value can always replicate the strategy of a firm with lower expected value), the difference is non-negative, and

$$\left|V_t^{\text{In}}(\hat{v}) - V_t^{\text{In}}(\tilde{v})\right| \leq \frac{\sup_{s \geq t} \pi_{\mathbb{J}s}}{\rho}\, |\hat{v} - \tilde{v}|.$$

Since $\sup_{s \geq t} \pi_{\mathbb{J}s} < \infty$, V_t^{In} is Lipschitz. □

Lemma 2. *If $\epsilon - 1 < 1/\varphi$, the platform problem* (12) *is concave and thus any solution is unique. If $\epsilon - 1 > 1/\varphi$, the platform's problem does not have a solution.*

Proof. Solving out the ODE for q_t yields

$$q_{kt} = e^{-\delta t} \int_0^t e^{\delta s} \ell_{ks}^{\varphi}\, \mathrm{d}s + e^{-\delta t} q_0.$$

The flow utility of platform k is then

$$\pi_{\mathbb{K}t} \frac{\left(e^{-\delta t} \int_0^t e^{\delta s} \ell_{ks}^{\varphi}\, \mathrm{d}s + e^{-\delta t} q_0\right)^{\epsilon - 1}}{\int_{\mathbb{K}} q_{zt}^{\epsilon-1}\, \mathrm{d}z} - \ell_{kt}.$$

Suppose that $\epsilon - 1 < 1/\varphi$. Then we observe that the flow utility must be concave in $\{\ell_{kt}\}$—The second term is linear; the first term's concavity is determined by the numerator, which is a CES aggregator with share weights determined by the exponential function. It is well known that this aggregator is strictly concave as long as $\epsilon - 1 < 1/\varphi$.

Since the flow utility is concave in $\{\ell_{kt}\}$ at each t, the objective function must also be concave in $\{\ell_{kt}\}$. As a result, any solution to platform k's problem must be unique.

Now suppose that $\epsilon - 1 > 1/\varphi$. I claim that there can not exist an equilibrium. Fix an arbitrary strategy $\{\ell_{kt}\}$ and then consider scaling up by some factor χ. For large χ, flow utility is determined primarily by the term

$$\chi^{\varphi(\epsilon-1)}\pi_{\mathbb{K}t}\frac{\left(e^{-\delta t}\int_0^t e^{\delta s}\ell_{ks}^{\varphi}\,\mathrm{d}s\right)^{\epsilon-1}}{\int_{\mathbb{K}} q_{zt}^{\epsilon-1}\,\mathrm{d}z} - \chi\ell_{kt}.$$

Thus if $\varphi(\epsilon-1) > 1$, it is possible for the platform to acheive an arbitrarily high value for the objective simply be scaling up χ. □

Lemma 3. *There exists a unique solution to the ODE system*

$$\begin{aligned}\dot{\ell}_{kt} &= \frac{\rho+\delta}{1-\varphi}\ell_{kt} - \frac{\varphi}{1-\varphi}\frac{\pi_{\mathbb{K}t}A(\epsilon-1)}{Kq_{kt}}\ell_{kt}^{\varphi} \\ \dot{q}_{kt} &= \ell_{kt}^{\varphi} - \delta q_{kt}\end{aligned} \tag{1}$$

for the whole domain $t \in [0,\infty)$ for any positive initial conditions (ℓ_{k0}, q_{k0}). Moreover, the solution for investment $\{\ell_{kt}\}$ is increasing and continuous in its initial condition ℓ_{k0}.

Proof. Since the system can be written in standard form as $[\dot{\ell}_{kt}\,\dot{q}_{kt}] = f(t, [\ell_{kt}\,q_{kt}])$ where f is continuous and locally Lipschitz in $[\ell_{kt}\,q_{kt}]$ the existence and uniqueness of local solutions follows from Picard-Lindelof. These properties also imply that solutions are continuous in initial conditions. Observe that the ODE for ℓ_{kt} has an absorbing point at zero and by Gronwall's inequality

$$\ell_{kt} \leq \ell_{k0}e^{\frac{\rho+\delta}{1-\varphi}t}$$

whenever a solution exists. Similarly from the ODE for q_{kt} we have

$$q_{kt} = e^{-\delta t}q_0 + \int_0^t e^{-\delta(t-s)}\ell_{ks}^{\varphi}\,\mathrm{d}s \leq e^{-\delta t}q_0 + \int_0^t e^{-\delta(t-s)}\ell_{k0}^{\varphi}e^{\varphi\frac{\rho+\delta}{1-\varphi}s}\,\mathrm{d}s.$$

But then the solution of the ODE system can not explode in finite time or become negative and so the maximal domain of existence must be all of $t \in [0,\infty)$.

To see that $\{\ell_{kt}\}$ is monotone in ℓ_{k0} divide both sides of the ODE for ℓ_{kt} by ℓ_{kt}. Then we have

$$\frac{\dot{\ell}_{kt}}{\ell_{kt}} = \frac{\rho+\delta}{1-\varphi} - \frac{\varphi}{1-\varphi}\ell_{kt}^{\varphi-1}\frac{\pi_{\mathbb{K}t}A(\epsilon-1)}{Kq_{kt}}.$$

Let $f_t = \ln(\ell_{kt})$. It suffices to show $\{f_t\}$ is monotone in f_0. We have

$$\dot{f}_t = \frac{\rho+\delta}{1-\varphi} - \frac{\varphi}{1-\varphi}(e^{f_t})^{\varphi-1}\frac{\pi_{\mathbb{K}t}A(\epsilon-1)}{Kq_{kt}}$$

with

$$\dot{q}_{kt} = e^{\varphi f_t} - \delta q_{kt}.$$

Now I observe that $\dot{f}$ is increasing in both f_t and q_{kt} with the former being true since $\varphi < 1$. Moreover, any solution for q_{kt} is monotone in $\{f_s, s \le t\}$.

Let $f_{10} > f_{20}$ be two initial conditions, with corresponding solutions $\{f_{1t}\}$ and $\{f_{2t}\}$, and suppose for contradiction that the ordering $f_{1t} > f_{2t}$ fails. By continuity there is a first time τ at which $f_{1\tau} = f_{2\tau}$, with $f_{1t} > f_{2t}$ on $[0,\tau)$. Since q_{kt} is strictly increasing in the history $\{f_s,\ s \le t\}$, the strict ordering on $[0,\tau)$ gives $q_{1\tau} > q_{2\tau}$. Evaluating $\dot{f}$ at τ, where $f_{1\tau} = f_{2\tau}$ so the dependence on f drops out, and using that $\dot{f}$ is strictly increasing in q, we obtain $\dot{f}_{1\tau} > \dot{f}_{2\tau}$. On the other hand, since $f_{1t} - f_{2t} > 0$ on $[0,\tau)$ and vanishes at τ, and the solutions are continuously differentiable, the gap is non-increasing at τ: $\dot{f}_{1\tau} - \dot{f}_{2\tau} \le 0$. This contradicts $\dot{f}_{1\tau} > \dot{f}_{2\tau}$. Therefore $f_{1t} > f_{2t}$ for all t, so $\{\ell_{kt}\}$ is increasing in ℓ_{k0}.

□

Lemma 4. *Any solution of the ODE system* (1) *for investment ℓ_{kt} for any initial conditions either diverges, vanishes, or converges to steady state.*

Proof. In what follows, let (ℓ_{SS}, q_{SS}) denote the unique steady state of the ODE system (1).

Step 1: If $q_{kt} \to q_{SS}$, then $\ell_{kt} \to \ell_{SS}$.

Suppose $q_{kt} \to q_{SS}$ and fix $\alpha > 0$. There exists $\zeta > 0$ sufficiently small such that

$$\frac{\rho+\delta}{1-\varphi} - \frac{\varphi}{1-\varphi}\frac{(\pi_{\mathbb{K}}+\zeta)\,A(\epsilon-1)}{K\,(q_{SS}-\zeta)}(\ell_{SS}+\alpha)^{\varphi-1} > 0. \tag{2}$$

Note that if α and ζ were zero, the left-hand side above would be zero, since (ℓ_{SS}, q_{SS}) is by definition a steady-state solution to (1).

Since $\pi_{\mathbb{K}t} \to \pi_{\mathbb{K}}$ and $q_{kt} \to q_{SS}$, for all t sufficiently large and any $\ell_{kt} \geq \ell_{SS} + \alpha$ we have

$$\begin{aligned}
\frac{\dot{\ell}_{kt}}{\ell_{kt}} &= \frac{\rho+\delta}{1-\varphi} - \frac{\varphi}{1-\varphi}\frac{\pi_{\mathbb{K}t}A(\epsilon-1)}{Kq_{kt}}\ell_{kt}^{\varphi-1} \\
&\geq \frac{\rho+\delta}{1-\varphi} - \frac{\varphi}{1-\varphi}\frac{(\pi_{\mathbb{K}}+\zeta)\,A(\epsilon-1)}{K\,(q_{SS}-\zeta)}(\ell_{SS}+\alpha)^{\varphi-1} \\
&> 0,
\end{aligned}$$

where the first inequality uses $\pi_{\mathbb{K}t} < \pi_{\mathbb{K}}+\zeta$, $q_{kt} > q_{SS}-\zeta$, and (since $\varphi-1 < 0$) $\ell_{kt}^{\varphi-1} \leq (\ell_{SS}+\alpha)^{\varphi-1}$ for $\ell_{kt} \geq \ell_{SS}+\alpha$.

Because $\ell^{\varphi-1}$ is decreasing, the inequality above shows that $\dot{\ell}_{kt}/\ell_{kt} > 0$ at every point with $\ell_{kt} \geq \ell_{SS}+\alpha$; hence the region $\{\ell \geq \ell_{SS}+\alpha\}$ is forward-invariant. Consequently, if $\ell_{kt} > \ell_{SS}+\alpha$ at some large t, then $\ell_{ks} > \ell_{SS}+\alpha$ for all $s \geq t$, the bound (2) applies throughout, and ℓ_{ks} diverges. But then $q_{kt} \to q_{SS}$ cannot hold. By the symmetric argument, $\dot{\ell}_{kt}/\ell_{kt} < 0$ whenever $\ell_{kt} \leq \ell_{SS}-\alpha$, so $\{\ell \leq \ell_{SS}-\alpha\}$ is forward-invariant; if $\ell_{kt} < \ell_{SS}-\alpha$ at some large t, then ℓ_{ks} vanishes and again $q_{kt} \to q_{SS}$ cannot hold. Therefore, for all large t, $\ell_{kt} \in [\ell_{SS}-\alpha, \ell_{SS}+\alpha]$. Since $\alpha > 0$ was arbitrary, $\ell_{kt} \to \ell_{SS}$.

Step 2: If $q_{kt} \nrightarrow q_{SS}$, then ℓ_{kt} diverges or vanishes.

Since $q_{kt} \nrightarrow q_{SS}$, there exists $\alpha > 0$ such that either $q_{kt} \geq q_{SS}+\alpha$ infinitely often or $q_{kt} \leq q_{SS}-\alpha$ infinitely often (or both). Fix such an α and take $\zeta > 0$ small and t large enough that $|\pi_{\mathbb{K}t} - \pi_{\mathbb{K}}| < \zeta$. We treat the two cases in turn.

1. **Case 1:** $q_{kt} \geq q_{SS}+\alpha$ **infinitely often.** Let ℓ^* solve $(\ell^*)^\varphi = \delta(q_{SS}+\alpha)$; this is the investment level that holds quality steady at $q_{SS}+\alpha$. Consider the first time τ at which $q_{k\tau} = q_{SS}+\alpha$. Since q reaches this level from below, $\dot{q}_{k\tau} \geq 0$, i.e. $\ell_{k\tau}^\varphi \geq \delta(q_{SS}+\alpha) = (\ell^*)^\varphi$, so $\ell_{k\tau} \geq \ell^*$.

   For ζ sufficiently small, at any point with $\ell \geq \ell^*$ and $q \geq q_{SS}+\alpha$,

$$\begin{aligned}
\frac{\dot{\ell}_{kt}}{\ell_{kt}} &= \frac{\rho+\delta}{1-\varphi} - \frac{\varphi}{1-\varphi}\frac{\pi_{\mathbb{K}t}A(\epsilon-1)}{Kq_{kt}}\ell_{kt}^{\varphi-1} \\
&> \frac{\rho+\delta}{1-\varphi} - \frac{\varphi}{1-\varphi}\frac{(\pi_{\mathbb{K}}+\zeta)A(\epsilon-1)}{K(q_{SS}+\alpha)}(\ell^*)^{\varphi-1} \\
&> 0,
\end{aligned}$$

   where the inequality uses $q_{kt} \geq q_{SS}+\alpha$ and, since $\varphi-1 < 0$, $\ell_{kt}^{\varphi-1} \leq (\ell^*)^{\varphi-1}$ for $\ell_{kt} \geq \ell^*$; both move the expression above its steady-state value of zero.

Two consequences follow. First, $\dot{\ell} > 0$ on this region, so ℓ cannot fall through ℓ^*. Second, $\ell \geq \ell^*$ gives $\dot{q} = \ell^\varphi - \delta q \geq 0$ at $q = q_{SS} + \alpha$, so q cannot fall through $q_{SS} + \alpha$. Hence the region $\{\ell \geq \ell^*,\, q \geq q_{SS} + \alpha\}$ is forward-invariant. The trajectory enters it at τ and remains, so the growth bound holds for all $s \geq \tau$, and $\ell_{ks} \to \infty$.

2. **Case 2:** $q_{kt} \leq q_{SS} - \alpha$ **infinitely often.** The argument mirrors Case 1. Let ℓ^{**} solve $(\ell^{**})^\varphi = \delta(q_{SS} - \alpha)$. At the first time q falls to $q_{SS} - \alpha$, q reaches the level from above, so $\dot{q} \leq 0$, giving $\ell^\varphi \leq \delta(q_{SS} - \alpha) = (\ell^{**})^\varphi$ and hence $\ell \leq \ell^{**}$. For ζ small, at any point with $\ell \leq \ell^{**}$ and $q \leq q_{SS} - \alpha$ the analogous bound gives $\dot{\ell}_{kt}/\ell_{kt} < 0$, so ℓ cannot rise through ℓ^{**}; and $\ell \leq \ell^{**}$ gives $\dot{q} \leq 0$ at $q = q_{SS} - \alpha$, so q cannot rise through $q_{SS} - \alpha$. Thus $\{\ell \leq \ell^{**},\, q \leq q_{SS} - \alpha\}$ is forward-invariant; the trajectory enters it and remains, and $\ell_{ks} \to 0$.

In either case ℓ_{kt} diverges or vanishes, completing the proof. □

Lemma 5. *There exists a unique solution to the boundary-value problem* (33).

Proof. Uniqueness. Suppose for contradiction there are two solutions ℓ_{kt} and $\hat{\ell}_{kt}$ with initial conditions ℓ_{k0} and $\hat{\ell}_{k0}$, and suppose $\hat{\ell}_{k0} > \ell_{k0}$. In the last part of the proof of Lemma 3 we showed that

$$\ln(\hat{\ell}_{kt}) - \ln(\ell_{kt}) \geq \ln(\hat{\ell}_{k0}) - \ln(\ell_{k0}) > 0$$

for all t. If both solutions converged to $\ell_{\mathbb{K}}$, this difference would tend to zero, contradicting the strictly positive lower bound. Hence at most one solution satisfies the terminal condition.

Existence. Consider a version of (33) in which the boundary at ∞ is replaced by a boundary at some finite $T > 0$. If a solution to this finite-horizon problem exists it is unique, by an argument analogous to the one just given. It is straightforward from (1) that increasing ℓ_{k0} makes ℓ_{kT} arbitrarily large, while decreasing ℓ_{k0} makes ℓ_{kT} arbitrarily close to zero: the right-hand side of (1) grows without bound as ℓ_{kt} diverges and is negative for all ℓ_{kt} in a neighborhood of zero. Thus, by continuity in the initial condition (Lemma 3), there is an initial condition, which we denote $\ell_{k0}(T)$, for which $\ell_{kT} = \ell_{SS}$; the corresponding solution to the initial value problem solves the boundary value problem with boundary at T.

Now take a sequence $\{T_n\}$ with $\lim_{n\to\infty} T_n = \infty$, and consider the corresponding solutions $\{\ell_{kt}(T_n)\}$, each extended to all t. By Lemma 4, each such solution diverges, vanishes, or converges to steady state. Hence there are either infinitely many that diverge or infinitely many that vanish.

Suppose first that infinitely many diverge. Define

$$\ell^*_{k0} = \inf\{\ell_{k0}(T_n) \mid \text{the extended solution with this initial condition diverges}\},$$

and let ℓ^*_{kt} be the solution with initial condition ℓ^*_{k0}. We claim ℓ^*_{kt} converges to steady state, and therefore solves the boundary value problem. Suppose not. By Lemma 4, ℓ^*_{kt} then either diverges or vanishes.

1. **Suppose ℓ^*_{kt} diverges.** Let $T^* = \sup\{t \mid \ell^*_{kt} = \ell_{SS}\}$. Since ℓ^*_{kt} diverges, it eventually exceeds ℓ_{SS} permanently, so $\{t \mid \ell^*_{kt} = \ell_{SS}\}$ is bounded above; hence if nonempty, T^* is finite. To see it is nonempty, suppose instead it is empty, so ℓ^*_{kt} never equals ℓ_{SS}; since it diverges, $\ell^*_{kt} > \ell_{SS}$ for all t. By monotonicity in the initial condition (Lemma 3), every solution with initial condition $\geq \ell^*_{k0}$ then also remains above ℓ_{SS} for all t. But each $\ell_{k0}(T_n)$ in the infimum defining ℓ^*_{k0} satisfies $\ell_{k0}(T_n) \geq \ell^*_{k0}$ and is, by construction, the initial condition of a solution with $\ell_{kT_n} = \ell_{SS}$ — a contradiction. Hence the set is nonempty and T^* is finite.

   Thus ℓ^*_{kt} solves the boundary value problem with boundary at T^*. Now choose $T_n > T^*$ such that the extended solution to the boundary problem at T_n diverges; this is possible since these solutions, being at ℓ_{SS} but generically not at q_{SS} at time T_n, diverge or vanish by Lemma 4, and infinitely many diverge by assumption. This solution reaches ℓ_{SS} at $T_n > T^*$, that is, later than ℓ^*_{kt} does. By monotonicity in the initial condition, a later crossing of ℓ_{SS} corresponds to a smaller initial condition, so $\ell_{k0}(T_n) < \ell^*_{k0}$. This contradicts $\ell^*_{k0} = \inf_n \ell_{k0}(T_n)$.

2. **Suppose ℓ^*_{kt} vanishes.** Inspecting (1), there exist $\underline{\ell} > 0$ and $\underline{q} > 0$ such that if $\ell_{kt} < \underline{\ell}$ and $q_{kt} < \underline{q}$ at any time, then ℓ_{kt} vanishes; that is, the region $\{\ell < \underline{\ell},\, q < \underline{q}\}$ is absorbing. Since ℓ^*_{kt} vanishes, fix $\alpha > 0$ and let T^* be the first time at which $\ell^*_{kT^*} < \underline{\ell} - \alpha$ and $q^*_{kT^*} < \underline{q} - \alpha$. Perturb the initial condition ℓ^*_{k0} upward by an arbitrarily small amount. By the definition of ℓ^*_{k0}, the perturbed solution diverges. Since any trajectory entering $\{\ell < \underline{\ell},\, q < \underline{q}\}$ vanishes, a diverging trajectory never enters this region; in particular, at the fixed time T^* the perturbed trajectory satisfies $\ell_{kT^*} \geq \underline{\ell}$ or $q_{kT^*} \geq \underline{q}$, whereas $\ell^*_{kT^*} < \underline{\ell} - \alpha$ and $q^*_{kT^*} < \underline{q} - \alpha$. This jump of at least α, produced by an arbitrarily small perturbation of the initial condition, contradicts continuity of solutions in the initial condition at the fixed finite time T^* (Lemma 3).

In both cases we reach a contradiction, so ℓ^*_{kt} converges to steady state and solves the boundary value problem.

If instead infinitely many of the extended solutions vanish, the symmetric argument applies, with ℓ^*_{k0} redefined as the supremum of the initial conditions whose extended solutions vanish; the analogous contradictions show ℓ^*_{kt} converges to steady state. Since one of the two cases must hold, existence follows. $\square$

C Supplemental Material for Section 5

Appendix C.1 proves Proposition 2. Appendix C.2 proves Proposition 3. Appendix C.3 presents additional comparative statics omitted from the main text.

C.1 Proof of Proposition 2

Proof of Proposition 2. The proof proceeds via two Lemmas.

Lemma 6 proves that product consumption is monotone in data informativeness.

Lemma 6. *An increase in G in the mean-preserving spread order leads to an increase in the steady state value of $M\mu_H$ and product consumption C and a decrease in the cdf H^c in second-order stochastic dominance.*

Proof. Suppose that G increases in the mean-preserving spread order to $\hat{G}$. Let $\hat{H}^c$ denote the steady state cdf under $\hat{G}$. Define

$$\gamma(y) = \hat{G}(y) - G(y) \qquad \text{and} \qquad \nu(y) = \hat{H}^c(y) - H^c(y).$$

By Part 3 of Theorem 1,

$$J\hat{G}(y) = M\hat{H}^c(y)^N + (J-M)\hat{H}^c(y)$$

and

$$JG(y) = MH^c(y)^N + (J-M)H^c(y).$$

Subtracting the second equation from the first gives

$$J\gamma(y) = M\left[\hat{H}^c(y)^N - H^c(y)^N\right] + (J-M)\nu(y).$$

Using the identity

$$a^N - b^N = (a-b)\sum_{k=0}^{N-1} a^{N-1-k}b^k,$$

it follows that

$$\gamma(y) = Q(y)\nu(y),$$

where

$$Q(y) = \frac{J-M}{J} + \frac{M}{J}\sum_{k=0}^{N-1} \hat{H}^c(y)^{N-1-k} H^c(y)^k.$$

Observe that Q is nondecreasing and satisfies

$$Q(y) \geq \frac{J-M}{J} > 0.$$

Define

$$V(s) = \int_0^s \nu(y)\,\mathrm{d}y.$$

Since $\hat{G}$ is a mean-preserving spread of G,

$$\int_0^s \gamma(y)\,\mathrm{d}y \geq 0$$

for every $s \geq 0$. Integrating by parts yields

$$Q(s)V(s) - \int_0^s V(y)\,\mathrm{d}Q(y) \geq 0. \tag{3}$$

We claim that $V(s) \geq 0$ for every $s \geq 0$. Suppose for contradiction that there exists $s_1 > 0$ such that

$$V(s_1) < 0.$$

Since V is continuous and $V(0) = 0$, there exists $s^* \in (0, s_1]$ at which V attains its minimum on $[0, s_1]$. Hence,

$$V(s^*) < 0 \qquad \text{and} \qquad V(y) \geq V(s^*)$$

for every $y \in [0, s^*]$. Since Q is nondecreasing,

$$\int_0^{s^*} V(y)\,\mathrm{d}Q(y) \geq V(s^*)\left[Q(s^*) - Q(0)\right].$$

Therefore, the left-hand side of (3) evaluated at s^* is at most

$$Q(s^*)V(s^*) - V(s^*)\left[Q(s^*) - Q(0)\right] = Q(0)V(s^*) < 0,$$

contradicting (3). Thus,

$$\int_0^s \nu(y)\,\mathrm{d}y = V(s) \geq 0$$

for every $s \geq 0$. Strict inequality holds for some s, since otherwise $\nu \equiv 0$, which together with $\gamma(y) = Q(y)\nu(y)$ implies $\gamma \equiv 0$, contradicting the assumption that $\hat{G}$ is a nontrivial mean-preserving spread of G.

Since $V \geq 0$, it follows that $\hat{H}^c$ is lower than H^c in second order stochastic dominance and thus $\mu_{\hat{H}^c} \leq \mu_{H^c}$. This in turn imples $\mu_{\hat{H}} > \mu_H$. □

Next, Lemma 2 shows that ad revenue is maximal in a limiting sense when data is uninformative. From Part 5 of Theorem 1, platform quality is then also maximal. To make the notation explicit, write $\pi_{\mathbb{K}}(G)$ for the average ad price as a function of the cdf G of expected values.

Lemma 7. *Let $\{G_n\}$ be a sequence of continuous cdfs converging pointwise to the Heaviside function centered at μ_F: $\lim_{n\to\infty} G_n(\hat{v}) = \mathbb{1}_{\{\hat{v}\geq\mu_F\}}$ for each $\hat{v} \in [0,\infty)$. Then $\lim_{n\to\infty} \pi_{\mathbb{K}}(G_n) = \sup_G \pi_{\mathbb{K}}(G)$ where the supremum is over all continuous cdfs G supported on $[0,\infty)$.*

Proof. Let G be arbitrary. We have

$$\begin{aligned}
\pi_{\mathbb{K}}(G) &= \mathbb{E}[B(\hat{v}^{(2)})] \\
&= \mathbb{E}\left[\pi_{\mathbb{J}} \int_0^{\hat{v}^{(2)}} \frac{1}{\rho + \lambda_f + \lambda_e(s)} \,\mathrm{d}s\right] \\
&\leq \frac{I\mathbb{E}[\hat{v}^{(2)}]}{\sigma M \mu_H(\rho + \lambda_f)} \\
&\leq \frac{I\mathbb{E}[\hat{v}^{(1)}]}{\sigma M \mu_H(\rho + \lambda_f)} \\
&= \frac{I}{\sigma M(\rho + \lambda_f)}
\end{aligned}$$

where above $\hat{v}^{(2)} \sim (H^c)^N + N\,(H^c)^{(N-1)}\,(1 - H^c)$ and $\hat{v}^{(1)} \sim (H^c)^N$. In the fourth line we use the fact that in steady state $H = (H^c)^N$. The notation has suppressed the dependency of H^c and H on G.

Using (17) in Theorem 1 we have

$$
\begin{aligned}
\lim_{n\to\infty} \pi_{\mathbb{K}}(G_n) &= \lim_{n\to\infty} \frac{I}{\sigma M \int_0^\infty [1 - H(s, G_n)]\, \mathrm{d}s} \\
&\quad \times \int_0^\infty \frac{1 - N H^c(s, G_n)^{N-1} + (N-1) H^c(s, G_n)^N}{\rho + \lambda_f + \lambda_a H^c(s, G_n)^{N-1}}\, \mathrm{d}s \\
&= \frac{I}{\sigma M \mu_F} \int_0^\infty \frac{1 - N \mathbb{1}_{s \geq \mu_F} + (N-1) \mathbb{1}_{s \geq \mu_F}}{\rho + \lambda_f + \lambda_a \mathbb{1}_{s \geq \mu_F}}\, \mathrm{d}s \\
&= \frac{I}{\sigma M \mu_F} \int_0^{\mu_F} \frac{1}{\rho + \lambda_f}\, \mathrm{d}s \\
&= \frac{I}{\sigma M (\rho + \lambda_f)}
\end{aligned}
$$

where in the second equality I have used the dominated convergence theorem to pass the limit through the integral. Above I have made explicit the dependency of H^c on G_n in the notation. $\square$

The proof of Proposition 2 is complete. $\square$

C.2 Proof of Proposition 3

Proof of Proposition 3. The proof proceeds via a series of lemmas.

Lemma 8. *An increase in ϵ leads to a decrease in the ad display rate A.*

Proof. By Theorem 1, A is equal to

$$\arg\max_a a\nu(a)^{\epsilon - 1} = \arg\max \ln(a) + (\epsilon - 1)\ln(\nu(a)). \tag{4}$$

Pick $\epsilon_2 > \epsilon_1$, and let a_1 solve (4) when $\epsilon = \epsilon_1$ and define a_2 analogously . By unique optimality,

$$\ln a_1 + (\epsilon_1 - 1) \ln \nu(a_1) > \ln a_2 + (\epsilon_1 - 1) \ln \nu(a_2)$$

and

$$\ln a_2 + (\epsilon_2 - 1) \ln \nu(a_2) > \ln a_1 + (\epsilon_2 - 1) \ln \nu(a_1).$$

Add these two equations and cancel terms:

$$(\epsilon_2 - \epsilon_1)\big(\ln \nu(a_2) - \ln \nu(a_1)\big) > 0.$$

Since $\epsilon_2 - \epsilon_1 > 0$ and $\ln \nu(\cdot)$ is decreasing, this implies $a_2 < a_1$. $\square$

Combining Lemma 8 and the following shows that an increase in ϵ leads to a decrease in product consumption.

Lemma 9. *Starting from steady state, an increase in A leads to a decrease in H_t and H_t^c in first-order stochastic dominance and an increase in $M_t\mu_{H_t}$ and C_{it} at all t.*

Proof. Recall from Step 3 of Appendix A.3 that

$$\dot{H}_t^c(\cdot) = \lambda_f \frac{JG(\cdot) - MH_t^c(\cdot)^N - (J-M)H_t^c(\cdot)}{J - M_t}.$$

Suppose the economy begins in the steady state associated with M_0. Then

$$JG(\cdot) - M_0 H_0^c(\cdot)^N - (J - M_0)H_0^c(\cdot) = 0.$$

Following an increase in A, we have $M > M_0$, so

$$\begin{aligned} &JG(\cdot) - MH_0^c(\cdot)^N - (J-M)H_0^c(\cdot) \\ &\quad = (M - M_0)\left[H_0^c(\cdot) - H_0^c(\cdot)^N\right] \geq 0. \end{aligned}$$

Moreover,

$$\frac{\partial}{\partial H_t^c}\left(JG - M(H_t^c)^N - (J-M)H_t^c\right) = -MN(H_t^c)^{N-1} - (J-M) < 0,$$

so the drift is strictly decreasing in H_t^c and has a unique zero, namely the new steady state. Since $J - M_t > 0$, the drift is positive whenever H_t^c lies below the new steady state and vanishes at the new steady state. Hence H_t^c increases pointwise over time following the increase in A. Equivalently, H_t^c decreases in the sense of first-order stochastic dominance.

Since

$$M_t\mu_{H_t} = J\mu_G - (J - M_t)\mu_{H_t^c},$$

and both M_t increases and $\mu_{H_t^c}$ decreases, it follows that $M_t\mu_{H_t}$ increases. By Part 7 of Theorem 1,

$$C_{it} = \frac{\sigma - 1}{\sigma} I (M_t\mu_{H_t})^{\frac{1}{\sigma-1}},$$

so C_{it} also increases.

Finally, recall that

$$\dot{H}_t = \frac{A}{M_t}\left[(H_t^c)^N - H_t\right].$$

Because the economy is initially in steady state,

$$H_0 = (H_0^c)^N.$$

Moreover, H_t^c is nondecreasing over time, so $(H_t^c)^N$ is a nondecreasing target. It follows from the differential equation that

$$H_t \leq (H_t^c)^N$$

at every t.

Now consider two post-change values $A' > A$. Since

$$M_t(A) = M_0 e^{-\lambda_f t} + \frac{A}{\lambda_f}(1 - e^{-\lambda_f t}),$$

we have

$$\frac{A}{M_t(A)} = \frac{A}{M_0 e^{-\lambda_f t} + \frac{A}{\lambda_f}(1 - e^{-\lambda_f t})},$$

which is increasing in A. We have also shown that

$$H_t^c(A') \geq H_t^c(A)$$

pointwise.

Suppose the two solutions for H_t coincide at some value h. Since $h \leq (H_t^c(A))^N$, their drifts satisfy

$$\begin{aligned} &\frac{A'}{M_t(A')}\left[(H_t^c(A'))^N - h\right] - \frac{A}{M_t(A)}\left[(H_t^c(A))^N - h\right] \\ &= \left[\frac{A'}{M_t(A')} - \frac{A}{M_t(A)}\right]\left[(H_t^c(A))^N - h\right] \\ &\quad + \frac{A'}{M_t(A')}\left[(H_t^c(A'))^N - (H_t^c(A))^N\right] \geq 0. \end{aligned}$$

A standard scalar comparison argument therefore implies that $H_t(A') \geq H_t(A)$ pointwise at every t. Thus H_t decreases in the sense of first-order stochastic dominance. $\square$

Combining Lemma 8 and the following lemma shows that an increase in ϵ leads to an increase in ad revenue $\pi_{\mathbb{K}} A$.

Lemma 10. *An increase in A leads to a decrease in steady state ad revenue $\pi_{\mathbb{K}} A$.*

Proof. From Part 4 of Theorem 1, we have

$$\pi_{\mathbb{K}} A = \frac{\lambda_f}{\sigma \int_0^\infty 1 - H^c(s)^N \, \mathrm{d}s} \int_0^\infty \frac{1 - NH^c(s)^{N-1} + (N-1)H^c(s)^N}{\rho + \lambda_f + \lambda_e(s)} \, \mathrm{d}s$$

where $\lambda_e(s) = \lambda_a H^c(s)^{N-1}$ for each $s \in [0, \infty)$.

To prove that ad revenue is decreasing, it suffices to prove the ratio of the integrand in the numerator to the integrand in the denominator is decreasing at each point s.

By Lemma 9, an increase in A leads to a decrease in H^c in first-order stochastic dominance. Thus, since λ_e increases pointwise, it suffices to show that

$$\frac{1 - NH^c(s)^{N-1} + (N-1)H^c(s)^N}{1 - H^c(s)^N} = N \frac{1 - H^c(s)^{N-1}}{1 - H^c(s)} - (N-1)$$

is decreasing in $H^c(s)$ which can be done simply by computing the derivative. I omit this step. □

This completes the proof of the proposition. □

C.3 Additional Comparative Statics Omitted From the Main Text

Lemma 11. *Starting from steady state, an increase in J leads to an increase in H_t^c and H_t in the sense of first-order stochastic dominance and thus to an increase in $M_t \mu_{H_t}$ and C_{it} at all t.*

Proof. Fix an argument of the cdfs and suppress it throughout. Recall that

$$\dot{H}_t^c = \lambda_f \frac{J(G - H_t^c) + M(H_t^c - H_t^{c\,N})}{J - M_t}.$$

Suppose the economy initially lies in the steady state associated with J_0. Then

$$J_0(G - H_0^c) + M(H_0^c - H_0^{c\,N}) = 0.$$

Following an increase from J_0 to $J > J_0$, the numerator of the post-change drift, evaluated at the initial steady state, is

$$J(G - H_0^c) + M(H_0^c - H_0^{c\,N}) = (J - J_0)(G - H_0^c).$$

Since

$$G - H_0^c = -\frac{M}{J_0} \left(H_0^c - H_0^{c\,N} \right) \leq 0,$$

it follows that the initial post-change drift satisfies

$$\dot{H}_0^c \leq 0.$$

Now define

$$Q_J(h) = J(G - h) + M(h - h^N).$$

Then

$$Q_J'(h) = -J + M\left(1 - Nh^{N-1}\right) \leq -J + M < 0,$$

so Q_J is strictly decreasing and has a unique zero, corresponding to the new steady state. Since $Q_J(H_0^c) \leq 0$, the solution decreases monotonically toward the new steady state. Hence H_t^c decreases pointwise. Equivalently, the distribution associated with H_t^c increases in the sense of first-order stochastic dominance.

Next, recall that

$$\dot{H}_t = \frac{A}{M_t}\left[(H_t^c)^N - H_t\right].$$

Neither A nor M_t changes with J. Moreover, because the economy begins in steady state,

$$H_0 = (H_0^c)^N.$$

Since H_t^c decreases pointwise, $(H_t^c)^N$ also decreases pointwise. Thus, at any point where two solutions for H_t coincide, the one associated with the larger value of J has a weakly smaller drift. A standard comparison argument for scalar differential equations therefore implies that H_t also decreases pointwise. Equivalently, the distribution associated with H_t increases in the sense of first-order stochastic dominance.

Since M_t is unaffected by J while μ_{H_t} increases, $M_t\mu_{H_t}$ increases. Finally, by Part 7 of Theorem 1,

$$C_{it} = \frac{\sigma - 1}{\sigma} I \left(M_t \mu_{H_t}\right)^{\frac{1}{\sigma-1}},$$

so C_{it} also increases. □

Lemma 12. *An increase in J leads to an increase in steady state ad revenue $\pi_{\mathbb{K}}A$.*

Proof. By Lemma 11, an increase in J leads to an increase in H^c in first-order stochastic dominance and then following the same steps as in Lemma 10, we see that this leads to an increase in steady state ad revenue $\pi_{\mathbb{K}}A$. □

Lemma 13. *An increase in ϵ leads to a decrease in C_{it} at each point in time and an increase in steady state ad revenue $\pi_{\mathbb{K}}A$ and if $K \leq 1$, steady state platform consumption X.*

Proof. An increase in ϵ leads to a decrease in A since (16) is submodular in ϵ and a_{kt}. By Lemma 10 this leads to an increase in steady state ad revenue $\pi_{\mathbb{K}}A$ and in turn platform investment (18) and thus platform quality. Since $X = K^{\frac{1}{\epsilon-1}}\nu(A)q_t$ and q_t increased while A decreased, if $K \leq 1$ then X must increase. □

Lemma 14. *An increase in K has no effects on ad revenue $\pi_{\mathbb{K}t}A$ at any time t and leads to an increase in steady state platform consumption X.*

Proof. As seen from (31), the equilibrium ad revenue $\pi_{\mathbb{K}t}A$ does not depend on K. In steady state, using (18)

$$X = K^{\frac{1}{\epsilon-1}}\frac{\ell_{\mathbb{K}}^{\varphi}}{\delta} = \frac{1}{\delta}K^{\frac{1}{\epsilon-1}-\varphi}\left(\frac{\varphi\delta\pi_{\mathbb{K}}A(\epsilon-1)}{(\delta+\rho)}\right)^{\varphi}.$$

This is increasing in K if the coefficient $\epsilon - 1 \leq 1/\varphi$. This is a necessary condition for an equilibrium to exist as shown in Lemma 2. □

D Supplemental Material for Section 6

Appendix D.1 presents proofs of the results in Section 6. Appendix D.2 presents general equilibrium analogs of Propositions 2 and 3.

D.1 Proofs of main results

Proof of Proposition 4. Product market clearing implies that $I_t = \frac{\sigma}{\sigma-1}(L - K\ell_{\mathbb{K}t})$. That is, income is equal to the markup times the quantity of labor left over for production.

To compute (19), let $\hat{\pi}_{\mathbb{K}} = \pi_{\mathbb{K}}/I$ be the average ad price per unit of income. From (17),

$$\hat{\pi}_{\mathbb{K}} = \frac{1}{\sigma M\mu_H}\int_0^{\infty}\frac{1 - NH^c(s)^{N-1} + (N-1)H^c(s)^N}{\rho+\lambda_f+\lambda_e(s)}\,\mathrm{d}s. \tag{5}$$

Then, rewriting (18), we have

$$\ell_{\mathbb{K}} = \frac{\varphi\delta\hat{\pi}_{\mathbb{K}}A(\epsilon-1)}{K\,(\rho+\delta)}I. \tag{6}$$

Given investment,

$$I = \frac{\sigma}{\sigma - 1}(L - K\ell_{\mathbb{K}}).$$

Solving this linear system for $\ell_{\mathbb{K}}$ yields (19). The rest of the proof is analogous to that of Theorem 1. □

Proof of Theorem 2. To characterize the planner's steady state investment, it suffices to take as given steady-state ad display A^* and suppose that we have already reached steady state for H_t. The Hamiltonian for the planner's problem for investment is

$$\mathcal{H}(t, q_t, \lambda_t, \ell_{\mathbb{K}t}) = \left[(L - K\ell_{\mathbb{K}t})\,(M\mu_H)^{\frac{1}{\sigma-1}}\right]^{1-\tau}\left[K^{\frac{1}{\epsilon-1}}\nu(A^*)q_t\right]^{\tau} + \lambda_t\left(\ell_{\mathbb{K}t}^{\varphi} - \delta q_t\right)$$

where λ_t satisfies

$$\rho\lambda_t - \dot{\lambda}_t = \left[(L - K\ell_{\mathbb{K}t})\,(M\mu_H)^{\frac{1}{\sigma-1}}\right]^{1-\tau}\tau K^{\frac{\tau}{\epsilon-1}}\nu(A^*)^{\tau}q_t^{\tau-1} - \delta\lambda_t.$$

Maximizing the Hamiltonian with respect to the control yields

$$\lambda_t\varphi\ell_{\mathbb{K}t}^{\varphi-1} = (1-\tau)K[L - K\ell_{\mathbb{K}t}]^{-\tau}\left[(M\mu_H)^{\frac{1}{\sigma-1}}\right]^{1-\tau}\left[K^{\frac{1}{\epsilon-1}}\nu(A^*)q_t\right]^{\tau}.$$

In steady state, λ_t must therefore be a constant. We have

$$\lambda_t = \frac{\left[(L - K\ell_{\mathbb{K}t})\,(M\mu_H)^{\frac{1}{\sigma-1}}\right]^{1-\tau}\tau K^{\frac{\tau}{\epsilon-1}}\nu(A^*)^{\tau}q_t^{\tau-1}}{\delta+\rho}.$$

Substituting ths into the previous equation, we have

$$\tau\frac{(L - K\ell_{\mathbb{K}t})\,\varphi\ell_{\mathbb{K}t}^{\varphi-1}}{\delta+\rho} = (1-\tau)Kq_t.$$

In steady state, $q_t = \ell_{\mathbb{K}t}^{\varphi}/\delta$ so then

$$\tau\frac{\varphi\,(L - K\ell_{\mathbb{K}t})}{\delta+\rho} = (1-\tau)K\frac{\ell_{\mathbb{K}t}}{\delta}.$$

Rearranging gives

$$\ell_{\mathbb{K}t} = \frac{\varphi\delta\frac{\tau}{1-\tau}}{\rho+\delta+\varphi\delta\frac{\tau}{1-\tau}}\frac{L}{K}.$$

The Hamiltonian is jointly concave in state and control and therefore satisfies the Mangasarian sufficient conditions for an optimal control.

To characterize the limiting ad display rate, let $U_t(a)$ denote flow utility under the constant ad display rate a, and let $U^{ss}(a)$ denote its steady-state value. Because $M_t\mu_{H_t}$ converges to its steady-state value at a rate independent of ρ, for every fixed ad display rate a,

$$\int_0^\infty e^{-\rho t} U_t(a)\,\mathrm{d}t = \frac{U^{ss}(a)}{\rho} + O(1),$$

as $\rho \to 0$, where the remainder is independent of ρ.

Suppose, by contradiction, that the limit of the planner's ad display rate does not maximize U^{ss}. Then some alternative $\tilde{a}$ yields a strictly larger steady-state flow payoff. By the expansion above, the transitional payoff difference is bounded, whereas the gain in discounted steady-state utility is of order $1/\rho$. Thus, for all sufficiently small ρ, switching to $\tilde{a}$ strictly increases welfare, contradicting optimality.

Since the planner's steady-state investment is independent of a, the terms in steady-state flow utility that depend on a are proportional to

$$[a\mu_H(a)]^{\frac{1-\tau}{\sigma-1}}\,\nu(a)^\tau.$$

Therefore, in the limit as $\rho \to 0$, the planner's ad display rate satisfies (23). □

Proofs of Proposition 5 and Corollary 2.2. These results follow immediately from Proposition 4 and Theorem 2.

□

D.2 Comparative Statics Omitted from the Main Text

Proposition 1. *The conclusions of Proposition 2 extend to the general equilibrium setting of Section 6.*

Proof. I first show that ad revenue is maximal in the limiting sense as data becomes uninformative. By an analogous argument to Lemma 2, $\hat{\pi}_{\mathbb{K}}$ is maximal as data becomes uninformative. Total ad revenue is $\pi_{\mathbb{K}}A = I\hat{\pi}_{\mathbb{K}}A$. Suppose for contradiction that ad revenue is not maximal in this limit; then under some informative data, I must be high enough that ad revenue rises. From the proof of Proposition 4, $I = \frac{\sigma}{\sigma-1}(L - K\ell_{\mathbb{K}})$, so a higher I requires lower investment $\ell_{\mathbb{K}}$. But (6) implies that lower investment requires lower ad revenue—a contradiction.

For product consumption, recall $C = \frac{\sigma-1}{\sigma} I (M\mu_H)^{1/(\sigma-1)}$. As data becomes uninformative, μ_H is minimal (Lemma 6). Ad revenue is maximal in this limit, so by (6) investment is maximal, which makes $I = \frac{\sigma}{\sigma-1}(L - K\ell_{\mathbb{K}})$ minimal. All three factors push C to its infimum.

Platform quality is maximal in the limit by Part 5 of Theorem 1, since investment is maximal. □

Proposition 2. *The conclusions of Proposition 3 on platform quality and product consumption extend to the general equilibrium setting, with $\hat{\pi}_{\mathbb{K}} A$ replacing $\pi_{\mathbb{K}} A$.*

Proof. By an analogous argument to that in the proof of Proposition 3, $\hat{\pi}_{\mathbb{K}} A$ rises with ϵ.

I next show that platform quality rises with ϵ. Suppose for contradiction that platform quality (and hence investment $\ell_{\mathbb{K}}$) falls when ϵ rises. By (6), lower investment implies lower ad revenue. Total ad revenue is $\pi_{\mathbb{K}} A = I \hat{\pi}_{\mathbb{K}} A$, so since $\hat{\pi}_{\mathbb{K}} A$ has risen, lower ad revenue requires lower I. But $I = \frac{\sigma}{\sigma-1}(L - K\ell_{\mathbb{K}})$, so lower I requires higher $\ell_{\mathbb{K}}$—a contradiction.

For product consumption, recall $C = \frac{\sigma-1}{\sigma} I (M\mu_H)^{1/(\sigma-1)}$. In general equilibrium, ϵ affects C only through I, since M and μ_H do not depend on ϵ. Since platform quality rises, investment $\ell_{\mathbb{K}}$ rises as well, so $I = \frac{\sigma}{\sigma-1}(L - K\ell_{\mathbb{K}})$ falls. Therefore C falls. □

E Supplemental Material for Section 7

This appendix presents supplemental material for the duopoly model of Section 7. Appendix E.1 provides details of the algorithm to compute an equilibrium. Appendix E.2 describes a reformulation in which there is a single state variable and uses it to reveal the connection between my model and that of Budd et al. (1993).

E.1 Details of the Algorithm to Compute Equilibrium

I use an implicit upwind finite difference scheme to solve (26). Here, I provide details of this finite-difference scheme. In particular, I describe the iterative procedure (used in Steps 1 and 2 outlined in Section 7) to calculate the value function at an earlier time step given the value function at a later time step. At the end of this subsection, I discuss properties of the algorithm and mathematical foundations.

Discretize the State Space I first compactify and then discretize each dimension of the state space in a standard way using evenly spaced grids. The two quality dimensions share a common grid with gap Δ_q, and the time dimension uses gap Δ_t between grid points.

Finite Difference Scheme Throughout, I supress the index k on the value function V_k for ease of notation. In the remaining parts of this document, let

$$V_{q_1}^{+}(q_1, q_2, t) \equiv \frac{V(q_1 + \Delta_q, q_2, t) - V(q_1, q_2, t)}{\Delta_q}$$

denote the forward difference and

$$V_{q_1}^{-}(q_1, q_2, t) \equiv \frac{V(q_1, q_2, t) - V(q_1 - \Delta_q, q_2, t)}{\Delta_q}$$

the backward difference approximation of $\partial V / \partial q_1$. Define V_t^+ analogously as the forward difference approximation of $\partial V / \partial t$.

Let

$$V_{q_1,q_1}^{c}(q_1, q_2, t) = \frac{V(q_1 + \Delta_q, q_2, t) - 2V(q_1, q_2, t) + V(q_1 - \Delta_q, q_2, t)}{\Delta_q^2}$$

denote the central difference approximation of $\partial^2 V / \partial q_1^2$. Define V_{q_2,q_2}^{c} analogously.

To implement the (semi-)implicit upwind scheme, I work with the following discretized version of (26):

$$\begin{aligned}
\rho V(q_1, q_2, t) = \sup_{\ell_{k1}} \, & V_t^{+}(q_1, q_2, t) + V_{q_1}^{+}(q_1, q_2, t)(\ell_{k1}^{\varphi} - \delta q_1)^{+} \\
& + V_{q_1}^{-}(q_1, q_2, t)(\ell_{k1}^{\varphi} - \delta q_1)^{-} + V_{q_2}^{+}(q_1, q_2, t)\,(\ell_{k2}^{\varphi} - \delta q_2)^{+} \\
& + V_{q_2}^{-}(q_1, q_2, t)\,(\ell_{k2}^{\varphi} - \delta q_2)^{-} - \ell_{k1} + \pi_{Kt} A \frac{q_1^{\epsilon-1}}{q_1^{\epsilon-1} + q_2^{\epsilon-1}}. \\
& + \frac{1}{2}\eta^2 q_1^2 \tilde{V}_{q_1,q_1}^{c}(q_1, q_2, t) + \frac{1}{2}\eta^2 q_2^2 \tilde{V}_{q_2,q_2}^{c}(q_1, q_2, t).
\end{aligned} \tag{7}$$

Thus, the finite-difference scheme is *upwind* in the sense that it uses the forward difference approximation of $\partial V / \partial q_1$ ($\partial V / \partial q_2$) whenever the drift of q_1 (q_2) is positive and the backward difference whenever the drift is negative. The scheme is *implicit* in the sense that it uses the forward-difference approximation of $\partial V / \partial t$. Note that (7) is a relationship between $V(\cdot, t + \Delta)$ and $V(\cdot, t)$ and thus we can use it to calculate $V(\cdot, t)$ given $V(\cdot, t + \Delta)$. The rest of this subsection details how this is done. I note that because the scheme is both upwind and implicit, $V(\cdot, t)$ depends *monotonically* on $V(\cdot, t+\Delta)$, which is essential for the algorithm's *stability*. Monotonicity and stability are critical for convergence of the numerical scheme to the solution (Barles and Souganidis, 1991).

Optimal Controls Using (7) to compute the value functions backward requires solving the optimization problem on the right-hand side. Using first-order conditions, we can show that there are only two candidates for the optimal controls: $\max\{\ell_k^+, \ell_k^c\}$ and $\min\{\ell_k^-, \ell_k^c\}$, where ℓ_k^+, ℓ_k^-, and ℓ_k^c are defined as follows.

Define ℓ_k^+ by

$$V_{q_1}^+(q_1, q_2, t)\varphi(\ell_k^+)^{\varphi-1} = 1,$$

ℓ_{k1}^- by

$$V_{q_1}^-(q_1, q_2, t)\varphi(\ell_k^-)^{\varphi-1} = 1,$$

and ℓ_k^c by

$$(\ell_k^c)^\varphi = \delta q_1.$$

Construct the M matrix Consider the following term which appears on the right-hand side of (26):

$$\begin{aligned} V_{q_1}^+(q_1, q_2, t)(\ell_{k1}^\varphi - \delta q_1)^+ &+ V_{q_1}^-(q_1, q_2, t)(\ell_{k1}^\varphi - \delta q_1)^- \\ &+ V_{q_2}^+(q_1, q_2, t)\,(\ell_{k2}^\varphi - \delta q_2)^+ + V_{q_2}^-(q_1, q_2, t)\,(\ell_{k2}^\varphi - \delta q_2)^- \\ &+ \frac{1}{2}\eta^2 q_1^2 V_{q_1,q_1}(q_1, q_2, t) + \frac{1}{2}\eta^2 q_2^2 V_{q_2,q_2}(q_1, q_2, t). \end{aligned} \tag{8}$$

I express the operator above in terms of a matrix M. Specifically, let N denote the number grid points for both quality dimensions. The expression (26) is to be expressed as MV where V is a $N^2 \times 1$ vector representing the value function at each quality (q_1, q_2) grid point and M is a $N^2 \times N^2$ matrix.

The rows of V are partitioned into N groups of N rows. Within each group, the state q_1 is the same and only q_2 varies.

More specifically, if[1]

$$V(q_1, q_2, t) \widehat{=} k$$

where k is the index of the vector V, then the corresponding indices of the other value function terms in (8) are:

$$V(q_1, q_2 + \Delta_q, t) \widehat{=} k + 1,$$

[1]Recall that $\widehat{=}$ means "corresponds."

$$V(q_1, q_2 - \Delta_q, t) \mathrel{\widehat{=}} k - 1,$$
$$V(q_1 + \Delta_q, q_2, t) \mathrel{\widehat{=}} k + N,$$
$$V(q_1 - \Delta_q, q_2, t) \mathrel{\widehat{=}} k - N.$$

Given the vector V, the rows and columns of M are organized accordingly. Namely, M is populated along the main diagonal (which contains the coefficients in (8) on $V(q_1, q_2, t)$), the +1 and -1 diagonals (which contain the coefficients on $V(q_1, q_2 + \Delta_q, t)$ and $V(q_1, q_2 - \Delta_q, t)$ respectively), and the $+N$ and $-N$ diagonals (contain the coefficients on $V(q_1+\Delta_q, q_2, t)$ and $V(q_1-\Delta_q, q_2, t)$) respectively. All other entries in M are zero. Thus, M is a *sparse matrix* with only 5 populated diagonals. This sparsity is *critical* for computational speed and a key advantage of the continuous-time formulation. I take advantage of it in the MATLAB code using the `spdiags` function.

Calculate Value Functions at Previous Time Step Using the M matrix (constructed using the *optimal controls at time t* in (8)) together with (7), we can succintly express $V(\cdot, t - \Delta_t)$ in terms of $V(\cdot, t)$ via the equation

$$V(\cdot, t - \Delta_t) = \left[I\left(1 + \rho\Delta_t\right) - \Delta_t M_t\right]^{-1}\left(\Delta_t g_t + V(\cdot, t)\right)$$

where I is the identity matrix and $g_t = -\ell^*_{kt} + \pi_{Kt} A \frac{q_1^{\epsilon-1}}{q_1^{\epsilon-1}+q_2^{\epsilon-1}}$ is the flow profit from the optimal control ℓ^*_{kt}. For clarity, I have explicitly indexed the M matrix and optimal control by time t.

Discussion of the Algorithm Though I have not proven convergence theoretically, the algorithm is informed by established mathematical foundations. The upwind finite-difference scheme used here falls within the class of monotone, stable, and consistent schemes that Barles and Souganidis (1991) prove converge to viscosity solutions of elliptic PDEs under suitable conditions. While their result does not directly apply to *coupled* HJB systems like the one studied here, the structural properties of the scheme offer reassurance about the reliability of the numerical method. The computation is also fast—even though there are two state variables, with minimal optimization and no parallelization, on a 2021 base model 16 inch M1 Macbook pro, the MATLAB code calculates steady state in roughly 2 minutes and the full equilibrium dynamics in roughly 10 minutes for reasonably fine time and quality grids.

Furthermore, the approach taken here has precedent in economics. For example, Brunnermeier and Sannikov (2017) use an analogous procedure to solve a coupled HJB system in a macrofinance model. Brunnermeier and

Sannikov (2017) consider a setting with a single state variable and atomistic agents, whereas I study a model with two state variables and atomic players. However, in both cases, computing equilibrium involves solving a coupled HJB system and the numerical algorithms we use are conceptually similar.

E.2 Model Variant with Single Spatial State Variable

Here I describe a slight reformulation of the duopoly model that reduces the number of state variables: instead of tracking the quality levels of both platforms separately, it suffices for each platform to track only the share of attention received by platform 1. In the process, I remark on the close connection between the duopoly model and that of Budd et al. (1993). As discussed in Section 7, this connection (together with the comment in Footnote 32) motivates the conjecture that a nontrivial amount of noise is sufficient for equilibrium existence in my setting. This is because Budd et al. 1993 show in their similar model that any nontrivial amount of noise ensures equilibrium existence (provided some standard and permissive conditions on other parameters are met).

Suppose that the effect of investment on quality scaled with quality. That is, suppose instead of (25) we have

$$\mathrm{d}q_{kt} = \left(\ell_{kt}^{\varphi} - \delta\right) q_{kt}\,\mathrm{d}t + \eta q_{kt}\,\mathrm{d}B_{kt}.$$

Then one can show that the share of attention x_{kt} that platform k receives is a Markov process. Namely, using Itô's formula,

$$\begin{aligned}\mathrm{d}x_{1t} = {} & x_{1t}(1-x_{1t})\left[(\epsilon-1)(\ell_{1t}^{\varphi}-\ell_{2t}^{\varphi}) + \frac{1}{2}(\epsilon-1)^2\eta^2(1-2x_{1t})\right]\mathrm{d}t \\ & + (\epsilon-1)\eta x_{1t}(1-x_{1t})\left(\mathrm{d}B_{1t}-\mathrm{d}B_{2t}\right).\end{aligned}$$

where recall that $x_{1t} = q_{1t}^{\epsilon-1}/(q_{1t}^{\epsilon-1}+q_{2t}^{\epsilon-1})$.

Therefore we can express

$$\begin{aligned}\mathrm{d}x_{1t} = {} & x_{1t}(1-x_{1t})\left[(\epsilon-1)(\ell_{1t}^{\varphi}-\ell_{2t}^{\varphi}) + \frac{1}{2}(\epsilon-1)^2\eta^2(1-2x_{1t})\right]\mathrm{d}t \\ & + (\epsilon-1)\eta x_{1t}(1-x_{1t})\sqrt{2}\,\mathrm{d}B_t.\end{aligned} \tag{9}$$

where $B \equiv \left(1/\sqrt{2}\right)(B_1 - B_2)$ is a standard Brownian motion.

Thus, in this setup it is logical to look for a steady state equilibrium that is Markov with only a single state variable: x_{1t}. That is, the two quality state variables collapse onto a single state variable.

The model in Budd et al. (1993) consists of a duopoly setting where profits depend on market share x_{kt} as in my model through some exogenously given function $\pi(\cdot)$. They do not focus on a specific microfoundation for market share and profits in terms of quality as I do but rather assume x_{kt} is directly controlled by the investment of the two players via the law of motion

$$\mathrm{d}x_{kt} = (\ell_{kt} - \ell_{-kt})\ \mathrm{d}t + \sigma\,\mathrm{d}B_t$$

where σ is a positive constant in the interior of the state space $[0, 1]$ and zero at the boundaries. Budd et al. (1993) also assume that there is some given flow cost $c(\cdot)$ of investment for both players.

Thus our models are qualitatively similar except for the fact that in (9) there is an extra factor $x_{1t}(1-x_{1t})$ that multiplies the drift and volatility terms and there is also the second term in brackets in the drift term.[2]

Note that at the boundaries of the state space, the volatility disappears in both of our models.

F Supplemental Material for Section 8

I present a characterization of the steady state equilibrium and a sketch of its derivation.

F.1 Steady State Equilibrium Characterization

The following Theorem 1 summarizes the steady state equilibrium properties.

Theorem 1. *Suppose that $\epsilon - 1 < 1/\varphi$ and that A is the unique solution of $\max_a a\nu(a)^{\epsilon-1}$. In any steady state equilibrium with increasing bidding functions the following hold:*

1. *Consumer i's demands for products are as in* (27) *and her demands for platforms are as in* (15).
2. *Firm j sets prices as in* (1).
3. *Platform k displays ads at rate A as in* (16).
4. *The size of consideration sets is $M = A/\lambda_f$.*

[2]Other differences are merely comestic such as the fact that investment is raised to the power φ in my model but not in Budd et al. (1993)—these differences disappear by appropriately relabeling variables.

5. *Firm j's expected flow profits from sales are as in* (2)*.*

6. *Bidding functions* $\boldsymbol{B} = (B_1, B_2)$ *for the two groups are the fixed point of the operator* $\boldsymbol{\Lambda}$ *defined by* (13) *which is a contraction map whenever values are bounded.*

7. *The attention shares received by the two groups are given by* (15)*.*

8. *The rates of investment by the two groups are given by* (14)*.*

Proof. I now sketch the proof, i.e., the procedure to solve for the steady state equilibrium. Much of the analysis of the baseline model ports over to this extended setup. Demands, prices, ad rates, and the size of consideration sets will be as in equations (27), (15), (16), and (29). However, we now must keep track of the joint distribution of the two signals inside and outside of consideration sets.

Step 1. Law of Motion of H_t and H_t^c: Let H_t^c denote the joint cdf of signals outside of consideration sets at time t. Let H_t denote the joint cdf of signals inside of consideration sets at time t. Let h_t^c and h_t be their corresponding pdfs. Let $H_t^c(\zeta_1, \infty)$ denote $\lim_{\zeta \to \infty} H_t^c(\zeta_1, \zeta)$ and let $H_t^c(\infty, \zeta_2)$ be defined analogously. Suppose that the winner in an auction on a platform in group z is the firm with the highest group z signal. Then, assuming M_t is at its steady state level, $M = A/\lambda_f$, the law of motion of h_t must satisfy

$$M \frac{\mathrm{d}}{\mathrm{d}t} \left[h_t(\boldsymbol{\zeta}) \right] = A \left[x_{1t} N H_t^c(\zeta_1, \infty)^{N-1} h_t^c(\boldsymbol{\zeta}) + x_{2t} N H_t^c(\infty, \zeta_2)^{N-1} h_t^c(\boldsymbol{\zeta}) - h_t(\boldsymbol{\zeta}) \right]. \quad (10)$$

In (10), with abuse of notation, x_{zt} denotes the total share of attention devoted to group z platforms. The first two terms in the brackets represents the inflow coming from the winners in the auctions on the two platform groups. The third term in the bracket represents the outflow as the consumer forgets about products.

To derive the steady state h, first fix an initial guess of x_1, the steady state level of x_{1t}. Then set $x_{1t} = x_1$, $x_{2t} = 1 - x_1$, and use the accounting identing $M_t h_t + (J - M_t) h_t^c = Jg$ to iterate (10) forward to convergence at each point $\boldsymbol{\zeta}$ in a fine grid on a region that contains almost all of G's mass.

Step 2. Calculate Bidding Strategies: Given M and h, we next compute equilibrium bidding strategies. To do so let

$$\mu_H = \int_{\mathbb{R}^2} \mathbb{E}[v_{ij}|\boldsymbol{\zeta}]h(\boldsymbol{\zeta})\,\mathrm{d}\boldsymbol{\zeta},$$

$$\pi_{\mathbb{J}} = \frac{I}{\sigma M \mu_H},$$

$$O_1(\cdot) = H^c(\cdot, \infty)^{N-1},$$

and

$$O_2(\cdot) = H^c(\infty, \cdot)^{N-1}.$$

Above, μ_H is the average value of the firms in consideration sets, $\pi_{\mathbb{J}}$ is the coefficient of firms' flow profits, O_1 determines the probability that a firm wins an auction if it takes place on a platform in group 1, and O_2 determines the probability that a firm wins an auction if it takes place on a platform in group 2.

In a steady state, bidding strategies correspond to a pair of functions $\boldsymbol{B} = (B_1, B_2)$. Here, $B_z : \mathbb{R} \mapsto [0, \infty)$ maps firm j's group z signal ζ_{zij} to its bid $B_z(\zeta_{zij})$ in a group z auction for consumer i. To derive $\boldsymbol{B}$, let $\boldsymbol{\zeta}_{ij} = (\zeta_{1ij}, \zeta_{2ij})$ and let $V(\boldsymbol{\zeta}_{ij})$ be firm j's continuation value from selling to consumer i at the time of auction entry if it knows $\boldsymbol{\zeta}_{ij}$ but does not know which platform hosts the auction.

More precisely, V satisfies the recursive equation

$$\begin{aligned} V(\boldsymbol{\zeta}_{ij}) = \sum_{z=1}^{2} x_z O_z(\zeta_{zij}) \left[\frac{\pi_{\mathbb{J}}}{\lambda_f + \rho} \mathbb{E}[v_{ij}|\boldsymbol{\zeta}_{ij}] + \frac{\lambda_f}{\lambda_f + \rho} \frac{\lambda_a}{\lambda_a + \rho} V(\boldsymbol{\zeta}_{ij}) \right] \\ - x_z \left[\int_{-\infty}^{\zeta_{zij}} B_z(s)\,\mathrm{d}O_z(s) + \left[1 - O_z(\zeta_{zij})\right] \frac{\lambda_a}{\lambda_a + \rho} V(\boldsymbol{\zeta}_{ij}) \right] \end{aligned} \tag{11}$$

The first term in the bracket is the discounted expected flow profit that firm j earns from entering Ω_{it}. It exits at rate λ_f and subsequently enters another auction at rate λ_a which corresponds to the second term in the bracket. On the second line, the first term in brackets is the expected payment in a group z auction. The last term is the continuation value in the event that firm j loses the auction, weighted by the probability that this happens.

Since auctions are second-price, in each auction, firm j simply bids the gain in its continuation value from winning the auction. Then,

$$
\begin{aligned}
B_z(\zeta_{zij}) &= \mathbb{E}\left[\frac{\pi_{\mathbb{J}}}{\lambda_f+\rho}v_{ij} + \frac{\lambda_f}{\lambda_f+\rho}\frac{\lambda_a}{\lambda_a+\rho}V(\boldsymbol{\zeta}_{ij})\middle|\zeta_{zij}, j \in \Omega_{it}^c\right] \\
&\quad - \mathbb{E}\left[\frac{\lambda_a}{\lambda_a+\rho}V(\boldsymbol{\zeta}_{ij})\middle|\zeta_{zij}, j \in \Omega_{it}^c\right] \\
&= \mathbb{E}\left[\frac{\pi_{\mathbb{J}}}{\lambda_f+\rho}v_{ij} - \frac{\rho}{\lambda_f+\rho}\frac{\lambda_a}{\lambda_a+\rho}V(\boldsymbol{\zeta}_{ij})\middle|\zeta_{zij}, j \in \Omega_{it}^c\right]. \qquad (12)
\end{aligned}
$$

Above, $\lambda_a = NA/(J-M)$ is the rate of auction entry. The expectation is conditional on only the group z signal and the fact that the firm is outside the consideration set since this is all that the firm knows when it bids.

Using (12) and (11), we can show that $\boldsymbol{B}$ is the fixed point of an operator $\boldsymbol{\Lambda} := (\Lambda_1, \Lambda_2)$ where $\Lambda_z : C^+(\mathbb{R})^2 \to C^+(\mathbb{R})$ takes in a pair of functions $\boldsymbol{f} = (f_1, f_2)$ and outputs another function[3]

$$
\Lambda_z(\boldsymbol{f})(\cdot) = \mathbb{E}\left[\frac{\pi_{\mathbb{J}}v_{ij} + \lambda_a \sum_{l=1}^{2} x_l \int_{-\infty}^{\zeta_{lij}} f_l(s)\,\mathrm{d}O_l(s)}{\lambda_f + \rho + \lambda_a \sum_{l=1}^{2} x_l O_l(\zeta_{zij})}\middle|\zeta_{zij} = \cdot, j \in \Omega_{it}^c\right]. \qquad (13)
$$

The map $\boldsymbol{\Lambda}$ is a contraction with modulus $\lambda_a/(\lambda_a + \lambda_f + \rho)$ with respect to the sup-norm whenever values v_{ij} are bounded above by some level $\overline{v}$. For numerical purposes, this will always be the case. Even without this bounded support assumption, we see that $\boldsymbol{\Lambda}$ is increasing. Thus, starting from an initial $\boldsymbol{B}$ such that $B_z > \Lambda_z(\boldsymbol{B})$ for each $z \in \{1,2\}$ it follows that $\{\boldsymbol{\Lambda}^n(\boldsymbol{B})\}_{n=1}^{\infty}$ is a decreasing sequence which converges to the fixed point. Thus we can compute the equilibrium bidding strategies by iterating on (13).

Step 3. Calculate Investment: In a steady state equilibrium, a platform in group $z \in \{1,2\}$ invests a constant level $\ell_{\mathbb{K}z}$ to maintain quality level $q_z = \ell_{\mathbb{K}z}^{\varphi}/\delta$.

Let platform k belong to group z. The Hamiltonian for platform k's optimization problem is

$$
\mathcal{H}(t, q_{kt}, \lambda_t, \ell_{kt}) = \pi_{\mathbb{K}z} A \frac{q_{kt}^{\epsilon-1}}{m_z q_z^{\epsilon-1} + m_{-z} q_{-z}^{\epsilon-1}} - \ell_{kt} + \lambda_t \left(\ell_{kt}^{\varphi} - \delta q_{kt}\right)
$$

where λ_t, the costate variable, evolves according to

$$
\rho\lambda_t - \dot{\lambda}_t = \pi_{\mathbb{K}z} A (\epsilon - 1) \frac{q_{kt}^{\epsilon-2}}{m_z q_z^{\epsilon-1} + m_{-z} q_{-z}^{\epsilon-1}} - \lambda_t \delta.
$$

[3] $C^+(\mathbb{R})$ denotes the set of nonnegative continuous functions on $\mathbb{R}$.

By the Maximum Principle, a necessary condition for optimality is that the control ℓ_{kt} maximizes the Hamiltonian along the optimal trajectory:

$$\lambda_t \varphi \ell_{kt}^{\varphi-1} = 1.$$

Under the conjectured stationary strategy then

$$\lambda_t \varphi \ell_{\mathbb{K}z}^{\varphi-1} = 1.$$

This implies that λ_t must be a constant λ. By the costate evolution equation,

$$\lambda = \frac{\pi_{\mathbb{K}z} A\left(\epsilon-1\right)}{\rho+\delta} \frac{q_z^{\epsilon-2}}{m_z q_z^{\epsilon-1} + m_{-z} q_{-z}^{\epsilon-1}}.$$

Substituting, we have

$$\frac{\pi_{\mathbb{K}z} A\left(\epsilon-1\right)}{\rho+\delta} \varphi \ell_{\mathbb{K}z}^{\varphi-1} = m_z q_z + m_{-z} \left(\frac{q_{-z}}{q_z}\right)^{\epsilon-2} q_{-z}.$$

This implies that

$$\frac{\pi_{\mathbb{K}z} A\left(\epsilon-1\right)}{\rho+\delta} \varphi \ell_{\mathbb{K}z}^{\varphi-1} = m_z \frac{\ell_{\mathbb{K}z}^{\varphi}}{\delta} + m_{-z} \left(\frac{\ell_{\mathbb{K}-z}}{\ell_{\mathbb{K}z}}\right)^{\varphi(\epsilon-2)} \frac{\ell_{\mathbb{K}-z}^{\varphi}}{\delta}.$$

Dividing both sides by $\ell_{\mathbb{K}z}^{\varphi}/\delta$ we arrive at

$$\frac{\delta \pi_{\mathbb{K}z} A\left(\epsilon-1\right)}{\rho+\delta} \varphi \ell_{\mathbb{K}z}^{-1} = m_z + m_{-z} \left(\frac{\ell_{\mathbb{K}-z}}{\ell_{\mathbb{K}z}}\right)^{\varphi(\epsilon-1)}.$$

By symmetry by considering the problem of a platform k in group $-z$,

$$\frac{\delta \pi_{\mathbb{K}-z} A\left(\epsilon-1\right)}{\rho+\delta} \varphi \ell_{\mathbb{K}-z}^{-1} = m_{-z} + m_z \left(\frac{\ell_{\mathbb{K}z}}{\ell_{\mathbb{K}-z}}\right)^{\varphi(\epsilon-1)}.$$

Let $y := \ell_{\mathbb{K}z}/\ell_{\mathbb{K}-z}$. Using the above two equations, I derive

$$\frac{\pi_{\mathbb{K}z}}{\pi_{\mathbb{K}-z}} \frac{1}{y} = \frac{m_z + m_{-z} y^{-\varphi(\epsilon-1)}}{m_{-z} + m_z y^{\varphi(\epsilon-1)}}.$$

Equivalently,

$$y = \left(\frac{\pi_{\mathbb{K}z}}{\pi_{\mathbb{K}-z}}\right)^{\frac{1}{1-\varphi(\epsilon-1)}}.$$

Thus,

$$\ell_{\mathbb{K}z} = \frac{\varphi\delta\pi_{\mathbb{K}z}A\left(\epsilon-1\right)}{\rho+\delta}\frac{1}{m_z + m_{-z}\left(\frac{\pi_{\mathbb{K}-z}}{\pi_{\mathbb{K}z}}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}}} \tag{14}$$

Note that if $\epsilon \leq 2$ then the Hamiltonian is jointly concave in the state and control and so I have identified the optimal control. If $\epsilon - 1 < \frac{1}{\varphi}$ one can verify using the same approach outline in Step 5 for the proof of Theorem 1.

Step 4. Calculate Quality Levels and Attention Shares: From (14) we also have the quality level

$$q_z = \frac{\ell_{\mathbb{K}z}^{\varphi}}{\delta}$$

and attention share

$$x_z = \frac{m_z q_z^{\epsilon-1}}{m_z q_z^{\epsilon-1} + m_{-z}q_{-z}^{\epsilon-1}} = \frac{m_z\left(\frac{\pi_{\mathbb{K}z}}{\pi_{\mathbb{K}-z}}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}}}{m_z\left(\frac{\pi_{\mathbb{K}z}}{\pi_{\mathbb{K}-z}}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}} + m_{-z}} \tag{15}$$

of group z platforms.

Step 5. Calculate Surpluses Consumer surplus is simply $u(C, X)/\rho$ where

$$C = \frac{\sigma-1}{\sigma}I(M\mu_H)^{\frac{1}{\sigma-1}}$$

and

$$X = \nu(A)\left(\sum_{z=1}^{2} m_z q_z^{\epsilon-1}\right)^{\frac{1}{\epsilon-1}}$$

are product and platform consumption respectively. □

F.2 Summary of the Algorithm

For ease of viewing, I summarize the algorithm once more here in the appendix.

1. Guess a value of x_1.
2. Iterate (10) forward to compute h.
3. Iterate (13) to compute per-unit income bid functions and average ad prices.

4. Check whether the guess of x_1 aligns with (15).

5. If yes, done. If not, repeat with a revised guess.

All other equilibrium objects are characterized in closed form in terms of the output of this algorithm and primitives. Though inefficient, one can simply run steps 2-4 for each guess of x_1 in a fine grid on $[0, 1]$. This is relatively fast and allows one to solve for *all* steady state equilibria and in particular, check uniqueness.

G Discussion of Dynamics

In this appendix, I describe some methodological advantages of a dynamic analysis.

Consider an alternative one period model in which firms participate in multiple auctions for a given consumer. In such a model, there will not exist a bidding equilibrium in symmetric strategies (and it is unclear if there are other asymmetric equilibria). It is easiest to show this when values are supported on a $[0, \overline{v}]$ where $\overline{v} < \infty$. Suppose that there was a symmetric bidding equilibrium with increasing bidding strategies. Then a firm with value $\overline{v}$ would have a profitable deviation to bidding a zero amount in one of the auctions it participates on. This is because the firm is guaranteed to win all of the auctions it participates on if it follows the equilibrium strategy. But, the firm has only unit demand for displaying an ad. By deviating in this way, the firm can reduce its cost while still displaying an ad. By spreading the auction competition out over time I am able to avoid this issue. Thus, dynamics allows us to have an auction analysis in which consumers multi-home and there is interplatform competition in the sale of ads.

Of course, we could consider a one period model where there is no interplatform competition in the sale of ads and each firm participates in one auction each. There would be some matching of firms to auctions which we would have to take a stance on. If N or A are sufficiently large, then regardless of the matching some firms *must* participate on multiple auctions—there aren't enough distinct bidders to be allocated to fill up the auctions. In other words, comparative statics on N or A would have to be limited to a certain range where this is not the case. This is unattractive for the model especially since A is endogenous. You could of course assume N and A are sufficiently small so that so that not all auctions are filled. But, these additional modeling assumptions, in my opinion, seem unnatural. An attempt to explain them would probably appeal to unmodeled frictions such as the fact that it takes time for firms to locate auctions for a given consumer. The dynamic modeling simply makes this intuition formal.

Consider an alternative one period formulation with competitive pricing in the ad market rather than auctions. The baseline model can be solved in a competitive pricing environment. However, in that model, ads would be sold consumer by consumer and it would be as though the platforms inform firms about the *identity* i of the consumer when they make their purchases. In reality, firms only see signals and they do not know if they correspond to the same individual just as they do in the baseline model of Section 3. Of course, a disadvantage of switching to competitive pricing is that future work can not explore the model using ad auction level data. Moreover, its not clear how to solve the extended version of the model where platforms may have different data with competitive pricing. In this model, we can not let platforms sell ads consumer by consumer as if the platforms know the consumers' identities because then the firms could combine the data they receive from the different platforms. Thus, suppose each firm sees only signals of the consumers' valuations but not their identities when choosing which ads to purchase. Firms would have to do some inference about the likelihood they will also purchase an ad for the same consumer on the other platforms. This inference effect leads to complicated purchasing strategies that are nontrivial to characterize.

H Extension: Network Effects

I extend the baseline model to allow for network effects.

H.1 Setup

I redefine the CES aggregate for platform consumption to be

$$X_{it} = \left[\int_{\mathbb{K}} [\eta(x_{kt})\nu(a_{kt})q_{kt}x_{ikt}]^{\frac{\epsilon-1}{\epsilon}} \, \mathrm{d}k \right]^{\frac{\epsilon}{\epsilon-1}}$$

where $\eta(x) = x^{\zeta}$ where $\zeta > 0$. I retain all other aspects of the baseline model of Section 3.

H.2 Equilibrium Characterization

Theorem 2. *Suppose that $\hat{A}$ is the unique solution of* $\max_a a\nu(a)^{\frac{\epsilon-1}{1-\zeta(\epsilon-1)}}$. *If $\hat{A}/\lambda_f < J$ and $0 < (\epsilon-1)/[1-\zeta(\epsilon-1)] < 1/\varphi$, then there exists a* *unique equilibrium in which each platform $k \in \mathbb{K}$ receives a positive amount of attention $x_{kt} > 0$ at all times t for any feasible initial conditions M_0, H_0, and q_0. The equilibrium converges to a steady state and has the following properties:*

1. *Consumer i's demands for products are as in (27) and her demands for platforms are as in (18).*

2. *Firm j sets prices as in (1).*

3. *Platform k displays ads at rate $\hat{A}$.*

4. *The size of consideration sets is given by (29) and the cdfs of the expected values of firms inside and outside of them are characterized by (30) and (4) with $\hat{A}$ in place of A.*

5. *Firm j's expected flow profits from sales are as in (2) and the rates at which firm j matches with consumers are as in (11).*

6. *Firm j bids according to (31).*

7. *Platform k's quality and investment solve the boundary-value problem (33) except with $(\epsilon - 1)/[1 - \zeta(\epsilon - 1)]$ in place of $\epsilon - 1$.*

8. *Total consumer, firm, and platform surplus are as in Step 8 of Section 4 except*
$$X_{it} = K^{\frac{1}{\epsilon-1} - \zeta} \nu(\hat{A}) q_t.$$

Moreover the sufficient conditions are almost necessary: if either $\hat{A}/\lambda_f \geq J$ or $(\epsilon - 1)/[1 - \zeta(\epsilon - 1)] > 1/\varphi$ then there does not exist an equilibrium.

Proof. As discussed in Section 9, the attention received by platform k solves

$$x_{kt}^{\zeta} = \frac{x_{kt}^{\zeta(\epsilon-1)} [\nu(a_{kt}) q_{kt}]^{\epsilon-1}}{Y} \tag{16}$$

where

$$Y = \int_{\mathbb{K}} x_{kt}^{\zeta(\epsilon-1)} [\nu(a_{kt}) q_{kt}]^{\epsilon-1} \, \mathrm{d}k.$$

Solving (16) for x_{kt} yields two possibilities:

$$x_{kt} = \frac{[\nu(a_{kt}) q_{kt}]^{\frac{\epsilon-1}{1-\zeta(\epsilon-1)}}}{Y^{\frac{1}{1-\zeta(\epsilon-1)}}} \tag{17}$$

or $x_{kt} = 0$. Under the equilibrium refinement, all platforms must receive positive attention share (17). Integrating both sides of (17) over $\mathbb{K}$ yields

$$Y^{\frac{1}{1-\zeta(\epsilon-1)}} = \int_{\mathbb{K}} [\nu(a_{kt})q_{kt}]^{\frac{\epsilon-1}{1-\zeta(\epsilon-1)}} \,\mathrm{d}k.$$

Then substituting into (17) gives

$$x_{kt} = \frac{[\nu(a_{kt})q_{kt}]^{\frac{\epsilon-1}{1-\zeta(\epsilon-1)}}}{\int_{\mathbb{K}} [\nu(a_{kt})q_{kt}]^{\frac{\epsilon-1}{1-\zeta(\epsilon-1)}} \,\mathrm{d}k}. \tag{18}$$

Thus, the only change relative to the baseline model is that the elasticity of attention with respect to platform quality is now higher. The rest of the equilibrium derivation follows the same steps as in Section 4. □

I Extension: Heterogeneous Platform Productivity

I solve an extension of the baseline model in which platforms may differ in the productivity of their investments.

I.1 Setup

Platform k now solves

$$\max_{\{\ell_{kt}\}} \int_0^\infty e^{-\rho t} \left(\pi_{\mathbb{K}t} A \frac{q_{kt}^{\epsilon-1}}{\int_{\mathbb{K}} q_{zt}^{\epsilon-1} \,\mathrm{d}z} - \alpha_k \ell_{kt} \right) \mathrm{d}t$$

where

$$\dot{q}_{kt} = \ell_{kt}^{\varphi} - \delta q_{kt}.$$

The only difference relative to the baseline model is the parameter $\alpha_k > 0$ which controls the productivity of platform k. Let P denote the frequency distribution of $\alpha_k, k \in \mathbb{K}$. I retain all other aspects of the baseline model.

I.2 Equilibrium Characterization

Theorem 3. *Suppose that A is the unique solution to*

$$\max_a a\nu(a)^{\epsilon-1},$$

that $\epsilon - 1 < 1/\varphi$, and that

$$\int \left(\frac{1}{\alpha}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}} \mathrm{d}P(\alpha) < \infty.$$

Suppose further that initial qualities satisfy

$$q_{k0} = \left(\frac{1}{\alpha_k}\right)^{\frac{\varphi}{1-\varphi(\epsilon-1)}} q_0.$$

Then there exists a unique equilibrium in which

$$\ell_{kt} = \left(\frac{1}{\alpha_k}\right)^{\frac{1}{1-\varphi(\epsilon-1)}} \ell_t \qquad \text{and} \qquad q_{kt} = \left(\frac{1}{\alpha_k}\right)^{\frac{\varphi}{1-\varphi(\epsilon-1)}} q_t,$$

where ℓ_t and q_t solve

$$(\rho+\delta)\frac{1}{\varphi}\ell_t - \frac{1-\varphi}{\varphi}\dot{\ell}_t = \frac{\pi_{\mathbb{K}t} A(\epsilon-1)}{\int \left(\frac{1}{\alpha}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}} \mathrm{d}P(\alpha)} \frac{\ell_t^{\varphi}}{q_t},$$
$$\dot{q}_t = \ell_t^{\varphi} - \delta q_t,$$

with initial condition q_0 and boundary condition

$$\lim_{t\to\infty} \ell_t = \frac{\delta \pi_{\mathbb{K}}^{ss} A(\epsilon-1)\varphi}{\rho+\delta} \left[\int \left(\frac{1}{\alpha}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}} \mathrm{d}P(\alpha)\right]^{-1},$$

where $\pi_{\mathbb{K}}^{ss} \equiv \lim_{t\to\infty} \pi_{\mathbb{K}t}$.

Proof. All equilibrium objects besides investment are derived as in the proof of Theorem 1.

The Hamiltonian for platform k's problem is

$$\mathcal{H}(t, q_{kt}, \ell_{kt}, \lambda_{kt}) = \pi_{\mathbb{K}t} A \frac{q_{kt}^{\epsilon-1}}{\int_{\mathbb{K}} q_{zt}^{\epsilon-1}\,\mathrm{d}z} - \alpha_k \ell_{kt} + \lambda_{kt}\left(\ell_{kt}^{\varphi} - \delta q_{kt}\right),$$

where the costate variable satisfies

$$\rho\lambda_{kt} - \dot{\lambda}_{kt} = \pi_{\mathbb{K}t} A(\epsilon-1) \frac{q_{kt}^{\epsilon-2}}{\int_{\mathbb{K}} q_{zt}^{\epsilon-1}\,\mathrm{d}z} - \delta\lambda_{kt}.$$

The first-order condition for maximizing the Hamiltonian yields

$$\lambda_{kt} = \frac{\alpha_k}{\varphi}\ell_{kt}^{1-\varphi}.$$

Differentiating with respect to time,

$$\dot{\lambda}_{kt} = \alpha_k \frac{1-\varphi}{\varphi}\ell_{kt}^{-\varphi}\dot{\ell}_{kt}.$$

Substituting into the costate equation gives

$$(\rho+\delta)\frac{\alpha_k}{\varphi}\ell_{kt}^{1-\varphi} - \alpha_k\frac{1-\varphi}{\varphi}\ell_{kt}^{-\varphi}\dot{\ell}_{kt} = \pi_{\mathbb{K}t}A(\epsilon-1)\frac{q_{kt}^{\epsilon-2}}{Q_t}, \tag{19}$$

where

$$Q_t = \int_{\mathbb{K}} q_{zt}^{\epsilon-1}\,\mathrm{d}z.$$

Step 1. Steady State. I first derive the steady-state equilibrium before solving for the transition path. In steady state,

$$(\rho+\delta)\frac{\alpha_k}{\varphi}\ell_k^{1-\varphi} = \pi_{\mathbb{K}}A(\epsilon-1)\frac{q_k^{\epsilon-2}}{Q},$$

and

$$q_k = \frac{\ell_k^{\varphi}}{\delta}.$$

Hence

$$(\rho+\delta)\frac{\alpha_k}{\varphi}\ell_k^{1-\varphi} = \pi_{\mathbb{K}}A(\epsilon-1)\frac{\ell_k^{(\epsilon-2)\varphi}}{\delta^{\epsilon-2}Q},$$

which implies

$$\ell_k = \left(\frac{\pi_{\mathbb{K}}A(\epsilon-1)}{\delta^{\epsilon-2}Q(\rho+\delta)}\frac{\varphi}{\alpha_k}\right)^{\frac{1}{1-\varphi(\epsilon-1)}}.$$

Therefore,

$$q_k = \frac{1}{\delta}\left(\frac{\pi_{\mathbb{K}}A(\epsilon-1)}{\delta^{\epsilon-2}Q(\rho+\delta)}\frac{\varphi}{\alpha_k}\right)^{\frac{\varphi}{1-\varphi(\epsilon-1)}}.$$

Using the definition of Q,

$$Q = \frac{1}{\delta^{\epsilon-1}}\int\left(\frac{\pi_{\mathbb{K}}A(\epsilon-1)}{\delta^{\epsilon-2}Q(\rho+\delta)}\frac{\varphi}{\alpha}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}}\,\mathrm{d}P(\alpha),$$

which implies

$$Q = \left[\frac{1}{\delta^{\epsilon-1}} \int \left(\frac{\pi_{\mathbb{K}} A(\epsilon-1)}{\delta^{\epsilon-2}(\rho+\delta)} \frac{\varphi}{\alpha} \right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}} \mathrm{d}P(\alpha) \right]^{1-\varphi(\epsilon-1)}.$$

Substituting back yields

$$\ell_k = \left(\frac{1}{\alpha_k} \right)^{\frac{1}{1-\varphi(\epsilon-1)}} \frac{\delta \pi_{\mathbb{K}} A(\epsilon-1)\varphi}{\rho+\delta} \left[\int \left(\frac{1}{\alpha} \right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}} \mathrm{d}P(\alpha) \right]^{-1}.$$

Step 2. Transition Path. Let ℓ_t and q_t denote the investment and quality of a platform with $\alpha_k = 1$. I conjecture that

$$\ell_{kt} = \left(\frac{1}{\alpha_k} \right)^{\frac{1}{1-\varphi(\epsilon-1)}} \ell_t$$

and

$$q_{kt} = \left(\frac{1}{\alpha_k} \right)^{\frac{\varphi}{1-\varphi(\epsilon-1)}} q_t.$$

The second relation is consistent with the law of motion because

$$\dot{q}_{kt} = \left(\frac{1}{\alpha_k} \right)^{\frac{\varphi}{1-\varphi(\epsilon-1)}} \dot{q}_t = \ell_{kt}^{\varphi} - \delta q_{kt},$$

provided that the initial conditions satisfy

$$q_{k0} = \left(\frac{1}{\alpha_k} \right)^{\frac{\varphi}{1-\varphi(\epsilon-1)}} q_0.$$

Substituting the conjecture into (19), the left-hand side becomes

$$\alpha_k \left(\frac{1}{\alpha_k} \right)^{\frac{1-\varphi}{1-\varphi(\epsilon-1)}} \left[(\rho+\delta) \frac{1}{\varphi} \ell_t^{1-\varphi} - \frac{1-\varphi}{\varphi} \ell_t^{-\varphi} \dot{\ell}_t \right].$$

Moreover,

$$q_{kt}^{\epsilon-2} = \left(\frac{1}{\alpha_k} \right)^{\frac{\varphi(\epsilon-2)}{1-\varphi(\epsilon-1)}} q_t^{\epsilon-2}.$$

The powers of α_k coincide because

$$1 - \frac{1-\varphi}{1-\varphi(\epsilon-1)} = -\frac{\varphi(\epsilon-2)}{1-\varphi(\epsilon-1)}.$$

Hence (19) reduces to

$$(\rho+\delta)\frac{1}{\varphi}\ell_t^{1-\varphi}-\frac{1-\varphi}{\varphi}\ell_t^{-\varphi}\dot{\ell}_t=\pi_{\mathbb{K}t}A(\epsilon-1)\frac{q_t^{\epsilon-2}}{Q_t}.$$

Under the conjectured quality profile,

$$Q_t=q_t^{\epsilon-1}\int\left(\frac{1}{\alpha}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}}\,\mathrm{d}P(\alpha).$$

Therefore,

$$(\rho+\delta)\frac{1}{\varphi}\ell_t^{1-\varphi}-\frac{1-\varphi}{\varphi}\ell_t^{-\varphi}\dot{\ell}_t=\frac{\pi_{\mathbb{K}t}A(\epsilon-1)}{\int\left(\frac{1}{\alpha}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}}\,\mathrm{d}P(\alpha)}\frac{1}{q_t}.$$

Multiplying both sides by ℓ_t^{φ} gives

$$(\rho+\delta)\frac{1}{\varphi}\ell_t-\frac{1-\varphi}{\varphi}\dot{\ell}_t=\frac{\pi_{\mathbb{K}t}A(\epsilon-1)}{\int\left(\frac{1}{\alpha}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}}\,\mathrm{d}P(\alpha)}\frac{\ell_t^{\varphi}}{q_t}.$$

Together with

$$\dot{q}_t=\ell_t^{\varphi}-\delta q_t,$$

this is exactly the reduced ODE system stated in the theorem.

Finally, Step 1 implies

$$\lim_{t\to\infty}\ell_t=\frac{\delta\pi_{\mathbb{K}}A(\epsilon-1)\varphi}{\rho+\delta}\left[\int\left(\frac{1}{\alpha}\right)^{\frac{\varphi(\epsilon-1)}{1-\varphi(\epsilon-1)}}\,\mathrm{d}P(\alpha)\right]^{-1}.$$

Existence and uniqueness of the solution to the reduced ODE system follow by the analogous arguments as in the proof of Theorem 1. Therefore, the conjectured scaling relations characterize the unique equilibrium. □

J Extension: Zero Prices

In this section, I give an informal argument that zero prices can arise, for some parameter conditions, in an equilibrium of a variant of the baseline model where platforms can charge nonnegative prices. I rule out negative prices on the grounds that it is too difficult for platforms to verify human usage as opposed to usage by bots. Under this premise, charging a negative price is unsustainable for a platform.

It is easiest to make the point when there are an integer number K of atomic platforms and when consumers have Cobb-Douglas utility as in Section 6. The core of the argument is simply that if consumers enjoy product consumption much more than platform consumption, then the attention spent on a platform will decrease quickly in the price set by the platform. This is because, if the consumer spends more attention on the platform, the consumer can spend less income on consuming products. When the elasticity of attention with respect to price is sufficiently high, it is better for a platform to rely solely on advertising to earn revenue.

Suppose that all platforms but platform 1 charge a zero price and have quality level q. Let q_1 denote platform 1's quality level. Let $p \geq 0$ denote the price charged by platform 1. Consumer i chooses how much attention x_1 to allocate to platform 1 to maximize flow utility which amounts to maximizing

$$(I - px_1)^{1-\tau} (M\mu_H)^{\frac{1-\tau}{\sigma-1}} \left[(x_1 q_1)^{\frac{\epsilon-1}{\epsilon}} + (K-1) \left(\frac{1-x_1}{K-1} q \right)^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\tau\epsilon}{\epsilon-1}} \nu(A)^\tau.$$

The first two terms comprise product consumption and the second two terms comprise platform consumption. Above, I have used the fact that the consumer will want to allocate attention evenly across the $K-1$ remaining platforms. After spending px_1 units of income on consuming platform 1, the consumer has only $I - px_1$ left to spend on products.

The first order condition for consumer i's problem is

$$\begin{aligned} p(1-\tau)(I-px_1)^{-\tau} &\left[(x_1 q_1)^{\frac{\epsilon-1}{\epsilon}} + (K-1) \left(\frac{1-x_1}{K-1} q \right)^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\tau\epsilon}{\epsilon-1}} = \\ &(I-px_1)^{1-\tau}\tau \left[(x_1 q_1)^{\frac{\epsilon-1}{\epsilon}} + (K-1) \left(\frac{1-x_1}{K-1} q \right)^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\tau\epsilon}{\epsilon-1}-1} \\ &\times \left[q_1 (x_1 q_1)^{\frac{\epsilon-1}{\epsilon}-1} - q (\frac{1-x_1}{K-1} q)^{\frac{\epsilon-1}{\epsilon}-1} \right]. \end{aligned}$$

Canceling out some terms and rearranging yields

$$p\left[(x_1q_1)^{\frac{\epsilon-1}{\epsilon}}+(K-1)\left(\frac{1-x_1}{K-1}q\right)^{\frac{\epsilon-1}{\epsilon}}\right]=$$
$$\frac{\tau}{1-\tau}\left[q_1(x_1q_1)^{\frac{\epsilon-1}{\epsilon}-1}-q\left(\frac{1-x_1}{K-1}q\right)^{\frac{\epsilon-1}{\epsilon}-1}\right](I-px_1)$$

which is linear in the price p.

Solving for p yields the inverse demand curve:[4]

$$p(x_1)=$$
$$\frac{I\left[q_1(x_1q_1)^{\frac{\epsilon-1}{\epsilon}-1}-q\left(\frac{1-x_1}{K-1}q\right)^{\frac{\epsilon-1}{\epsilon}-1}\right]}{\frac{1-\tau}{\tau}\left[(x_1q_1)^{\frac{\epsilon-1}{\epsilon}}+(K-1)\left(\frac{1-x_1}{K-1}q\right)^{\frac{\epsilon-1}{\epsilon}}\right]+\left[q_1(x_1q_1)^{\frac{\epsilon-1}{\epsilon}-1}-q\left(\frac{1-x_1}{K-1}q\right)^{\frac{\epsilon-1}{\epsilon}-1}\right]x_1}.$$

Since $I-px_1$ must be positive and p must be nonnegative, the domain of the inverse demand curve is the set of demands x_1 such that the numerator is nonnegative. That is, the domain is

$$x_1\in\left[0,\frac{q_1^{\epsilon-1}}{q_1^{\epsilon-1}+(K-1)q^{\epsilon-1}}\right].$$

This is intuitive: the domain consists of attention levels that are less than that which would arise if platform 1 also charged a price of zero.

We can now formulate platform 1's pricing problem which is to choose x_1 in this domain to maximize flow profit:

$$p(x_1)x_1+\pi_{\mathbb{K}}Ax_1.$$

We are interested in parameter conditions for when

$$x_1=\frac{q_1^{\epsilon-1}}{q_1^{\epsilon-1}+(K-1)q^{\epsilon-1}}$$

is optimal which corresponds to setting a zero price. This amounts to looking for parameter conditions such that

$$\frac{\mathrm{d}[p(x_1)x_1]}{\mathrm{d}x_1}+\pi_{\mathbb{K}}A=p'(x_1)x_1+p(x_1)+\pi_{\mathbb{K}}A>0 \tag{20}$$

[4] One can see that the inverse demand curve is monotone since the objective is submodular in (p,x_1).

for all x_1 in the domain. This will happen if $p(\cdot)$ does not decrease too fast. Intuitively this will be the case when τ is close to zero so that the consumer cares little about platform use and so attention will be very sensitive to the price set by platform 1.

By inspection, if τ is sufficiently close to 0,

$$p(x_1) \approx I \frac{\tau}{1-\tau} \frac{q_1 (x_1 q_1)^{\frac{\epsilon-1}{\epsilon}-1} - q \left(\frac{1-x_1}{K-1} q\right)^{\frac{\epsilon-1}{\epsilon}-1}}{(x_1 q_1)^{\frac{\epsilon-1}{\epsilon}} + (K-1) \left(\frac{1-x_1}{K-1} q\right)^{\frac{\epsilon-1}{\epsilon}}}.$$

I show later that we can bound

$$\frac{\mathrm{d}[p(x_1) x_1]}{\mathrm{d}x_1}$$

from *below* for all x_1 in the domain by an amount that can be made arbitrarily close to 0 by making τ sufficiently close to 0. Thus, when τ is sufficiently close to 0, it follows that (20) holds for all x_1 in the domain and a price of zero is optimal. I will take this fact as given now and show it formally later.

I have therefore shown that platform 1 does not have a profitable deviation to charging a positive price when τ is close to 0. In principle the platform could deviate both in its investment strategy and in its pricing strategy. But, so far, our analysis has fixed an arbitrary quality level for q_1. For some value of τ, say $\tau(q_1)$, which depends on q_1 we have shown it is not profitable to charge a positive price. Let $\overline{q}$ be relatively large and $\underline{q}$ be relatively small and consider

$$\tau^* = \inf_{q_1 \in [\underline{q}, \overline{q}]} \tau(q_1) < 1.$$

Then for parameter $\tau = \tau^*$ it is never optimal for a platform to deviate to any quality $q \in [0, \overline{q}]$. By setting $\overline{q}$ sufficiently high, investment costs are sufficiently high that it is obviously not optimal to deviate in terms of investment to end up at any quality $q \geq \overline{q}$. Similarly if $q \leq \underline{q}$ for some $\underline{q}$ sufficiently low deviating by cutting back on investment is not profitable because profits are too low at any positive price that the platform can set.

I will now show that we can bound the derivative of

$$p(x_1) x_1 = \frac{I \left[q_1 (x_1 q_1)^{\frac{\epsilon-1}{\epsilon}-1} - q \left(\frac{1-x_1}{K-1} q\right)^{\frac{\epsilon-1}{\epsilon}-1} \right] x_1}{\frac{1-\tau}{\tau} \left[(x_1 q_1)^{\frac{\epsilon-1}{\epsilon}} + (K-1) \left(\frac{1-x_1}{K-1} q\right)^{\frac{\epsilon-1}{\epsilon}} \right] + \left[q_1 (x_1 q_1)^{\frac{\epsilon-1}{\epsilon}-1} - q \left(\frac{1-x_1}{K-1} q\right)^{\frac{\epsilon-1}{\epsilon}-1} \right] x_1}$$

from below by an amount arbitrarily close to 0 as claimed.

Define f such that

$$p(x_1)x_1 = I\frac{f'(x_1)x_1}{\frac{1-\tau}{\tau}\frac{\epsilon-1}{\epsilon}f(x_1)+f'(x_1)x_1}.$$

Then

$$\begin{aligned}\frac{1}{I}\frac{\mathrm{d}[p(x_1)x_1]}{\mathrm{d}x_1} &= \frac{f''(x_1)x_1+f'(x_1)}{\frac{1-\tau}{\tau}\frac{\epsilon-1}{\epsilon}f(x_1)+f'(x_1)x_1} \\ &\quad - \frac{f'(x_1)x_1}{\left[\frac{1-\tau}{\tau}\frac{\epsilon-1}{\epsilon}f(x_1)+f'(x_1)x_1\right]^2}\left[\left(\frac{1-\tau}{\tau}\frac{\epsilon-1}{\epsilon}+1\right)f'(x_1)+f''(x_1)x_1\right].\end{aligned}$$

Can we bound this from below? Consider taking the limit as τ tends to 0. Then the above, for a fixed x_1, converges to 0. However, we need to make sure that the infimum of the above over the entire range converges to 0. This will necessarily be the case if we can bound $f'(x_1)$, $f''(x_1)$, $f(x_1)$, and $f'(x_1)x_1$. The concern is about points x_1 near 0 since some of these terms explode there. Suppose we know that for any τ close to 0 that it is not optimal to set $x_1 < \tilde{\epsilon}$ for some $\tilde{\epsilon} > 0$, then we are done since $f'(x_1)$, $f''(x_1)$, $f(x_1)$, and $f'(x_1)x_1$ are bounded on $\left[\tilde{\epsilon}, \frac{q_1^{\epsilon-1}}{q_1^{\epsilon-1}+(K-1)q^{\epsilon-1}}\right]$

To show this, note that

$$p(x)x + \pi_{\mathbb{K}}Ax \leq p(x)x + \pi_{\mathbb{K}}A\tilde{\epsilon}$$

whenever $x \leq \tilde{\epsilon}$. But

$$p(x)x \leq I\frac{f'(x)x}{\frac{\epsilon-1}{\epsilon}f(x)+f'(x)x}$$

for all $\tau \leq 1/2$. Then

$$p(x)x + \pi_{\mathbb{K}}Ax \leq \sup_{x\in[0,\tilde{\epsilon}]} I\frac{f'(x)x}{\frac{\epsilon-1}{\epsilon}f(x)+f'(x)x} + \pi_{\mathbb{K}}A\tilde{\epsilon}$$

and the upper bound holds uniformly over all $\tau \leq \frac{1}{2}$. For some $\tilde{\epsilon} > 0$ this right hand side is always less than

$$\pi_{\mathbb{K}}A\frac{q_1}{q_1+(K-1)q}.$$

Thus, there is some $\tilde{\epsilon} > 0$ lower bound such that its not optimal to set $x_1 < \tilde{\epsilon}$ for any parameter $\tau \leq 1/2$.

K Extension: Firm and Platform Entry

I extend the baseline model to allow for entry of firms and platforms.

K.1 Setup

To enter the market, a firm must pay a cost $e_{\mathbb{J}} > 0$ and a platform must pay a cost $e_{\mathbb{K}} > 0$. I retain all other aspects of the baseline model of Section 3 except that in equilibrium, the measure J of firms and the measure K of platforms are such that firms and platforms earn zero profits net of entry costs.

K.2 Steady-State Equilibrium Characterization

For the notion of steady state equilibrium, I assume that each platform enters with the steady state quality level to keep the analysis simple. One might consider other conventions such as having a given platform enter with some given quality level q_0 that may differ from steady state and then characterize transition dynamics for that platform while restricting all other equilibrium properties in steady state. I will not explore that here. Under either convention, the measure of firms in the steady state will be the same.

Theorem 4. *Suppose that A is the unique solution of $\max_a a\nu(a)^{\epsilon-1}$. If $A/\lambda_f < J$ and $\epsilon \leq 1/\varphi$ then there exists a steady state equilibrium. If one exists, it is unique and has the same steady-state properties as in Theorem 1 for a given J and K which satisfy*

$$K = \frac{1}{\rho}\frac{\pi_{\mathbb{K}} A}{e_{\mathbb{K}}}\left(1 - \frac{\varphi\delta(\epsilon-1)}{\rho+\delta}\right), \tag{21}$$

and

$$J = \frac{1}{\rho}\left(\frac{\frac{I}{\sigma} - \pi_{\mathbb{K}} A}{e_{\mathbb{J}}}\right). \tag{22}$$

Proof. Equations 21 and (22) are zero profit conditions. Note that in (22), $\pi_{\mathbb{K}}$ depends on J. Thus, to prove uniqueness I must prove there is a unique solution for J in (22). This follows by Lemma 12 which shows that $\pi_{\mathbb{K}}$ is increasing in J. The other parts of the theorem follows the same roadmap as in the proof of Theorem 1. □

It is straightforward to extend most of the comparative statics for steady state in Appendix C to this setting.

Proposition 3. *In steady state, banning data use in the sense of Proposition 2 leads to a decrease in J and an increase in K.*

Proof. Suppose for contradiction J increases. Holding fixed J, a ban on data use raises ad revenue, as show in Proposition 2 of the main text. An increase in J further increases ad revenue as shown by Lemma 12. Thus ad revenue must increase. But then by (22), J must decrease, a contradiction. Thus, J must decrease.

Because ad revenue must increase, inspecting (21), reveals K must increase. $\square$

Proposition 4. *In steady state, an increase in platform substitutability ϵ leads to a decrease in J and, if the induced decrease in the ad display rate A is sufficiently small, leads to a decrease in K.*

Proof. An increase in ϵ leads to a decrease in A, which increases ad revenue by Lemma 10. Suppose, toward a contradiction, that J increases. By Lemma 12, the increase in J further increases ad revenue. Hence, ad revenue must increase. Equation (22) then implies that J decreases, a contradiction. Thus, J decreases.

If the decrease in A induced by the increase in ϵ is sufficiently small, then (21) implies that K decreases. $\square$

L Extension: Reserve Prices

I extend the baseline model by allowing platforms to set reserve prices in their advertising auctions.

L.1 Setup

Each platform chooses both an auction-opportunity rate and a reserve price. An auction opportunity temporarily interrupts the consumer's use of the platform and therefore generates nuisance costs whether or not the reserve price is met (i.e, even if an ad is not allocated the resulting empty ad slot still generates nuisance costs). If no bidder submits a bid above the reserve price, no ad is displayed and no firm enters the consumer's consideration set.

All other aspects of the model are as in Section 3.

L.2 Steady-State Equilibrium Characterization

Define the virtual value associated with the distribution H^c by

$$\psi^c(v) = v - \frac{1 - H^c(v)}{h^c(v)}.$$

The following theorem derives a system of equilibrium conditions for the reserve-price extension. It shows that any regular solution to this system induces a steady-state equilibrium.

Theorem 5. *Suppose that $\epsilon - 1 < 1/\varphi$ and that*

$$A = \arg\max_{a \geq 0} a\nu(a)^{\epsilon-1}.$$

If there exists a triple (M, Y, H^c) satisfying the equilibrium conditions in items 4, 6, and 8, where H^c is continuously differentiable with strictly positive density h^c and ψ^c is strictly increasing, then this triple induces a steady-state equilibrium with the following properties:

1. *Consumer i's demands for products are as in* (27) *and her demands for platforms are as in* (15)*.*

2. *Firm j sets prices as in* (1)*.*

3. *Each platform holds auctions at rate*

$$A = \arg\max_{a \geq 0} a\nu(a)^{\epsilon-1}.$$

4. *The rate at which ads are successfully displayed is*

$$A\left[1 - H^c(Y)^N\right],$$

and the mass of firms in a consumer's consideration set is

$$M = \frac{A\left[1 - H^c(Y)^N\right]}{\lambda_f}.$$

5. *Firm j's expected flow profits from sales are as in* (2)*.*

6. *Each platform sets reserve price*

$$R = \frac{\pi_{\mathbb{J}}}{\lambda_f + \rho} Y,$$

where Y solves

$$Y = \frac{1 - H^c(Y)}{h^c(Y)}.$$

7. Firms with expected value $\hat{v}_{ij} \geq Y$ bid according to

$$B(\hat{v}_{ij}) = R + \pi_{\mathbb{J}} \int_{Y}^{\hat{v}_{ij}} \frac{1}{\rho + \lambda_f + \lambda_a H^c(s)^{N-1}}\, ds,$$

where

$$\lambda_a = \frac{NA}{J - M}.$$

8. The distribution H^c satisfies

$$H^c(s) = \frac{J}{J - M} G(s), \qquad s \leq Y,$$

and

$$H^c(s)^N - H^c(Y)^N = \left[\frac{J}{M} G(s) - \frac{J - M}{M} H^c(s)\right]\left[1 - H^c(Y)^N\right], \qquad s \geq Y.$$

9. Each platform invests at the rate characterized in (18)*.*

Proof. Consumer demand, firms' product prices, and firms' flow profits from product sales are determined as in the baseline model.

An auction opportunity generates nuisance costs regardless of whether a bid clears the reserve price. If a platform holds auctions at rate a, its flow advertising revenue is proportional to a, while its demand is proportional to $\nu(a)^{\epsilon-1}$. Hence each platform chooses

$$A = \arg\max_{a \geq 0} a\nu(a)^{\epsilon-1}.$$

An auction results in an ad display if and only if at least one of the N bidders has a value above Y. The probability of this event is

$$1 - H^c(Y)^N.$$

Consequently, ads are displayed at rate

$$A\left[1 - H^c(Y)^N\right].$$

Matching the inflow of firms into consideration sets with forgetting gives

$$\lambda_f M = A\left[1 - H^c(Y)^N\right],$$

and therefore

$$M = \frac{A\left[1 - H^c(Y)^N\right]}{\lambda_f}.$$

Because each auction has N bidders selected from the $J-M$ mas sof firms outside the consumer's consideration set, a given outside firm enters auctions at rate

$$\lambda_a = \frac{NA}{J-M}.$$

In a second-price auction, a firm bids the gain its continuation value from winning the auction. Thus

$$B(\hat{v}_{ij}) = \frac{\pi_{\mathbb{J}}}{\lambda_f + \rho}\hat{v}_{ij} + \frac{\lambda_f}{\lambda_f + \rho}\frac{\lambda_a}{\lambda_a + \rho}V(\hat{v}_{ij}) - \frac{\lambda_a}{\lambda_a + \rho}V(\hat{v}_{ij}) \tag{23}$$

where $V(\hat{v}_{ij})$ is the continuation value from selling to consumer i in the future, at the time of entry into an auction. It is defined recursively by the equation

$$\begin{aligned} V(\hat{v}_{ij}) = \left[1 - H(\hat{v}_{ij})^{N-1}\right] \frac{\lambda_a}{\lambda_a + \rho}V(\hat{v}_{ij}) + \\ H^c(\hat{v}_{ij})^{N-1}\Bigg(\frac{\pi_{\mathbb{J}}}{\lambda_f + \rho}\hat{v}_{ij} + \frac{\lambda_f}{\lambda_f + \rho}\frac{\lambda_a}{\lambda_a + \rho}V(\hat{v}_{ij}) \\ - \mathbb{E}\left[\max\{B(\hat{v}^{(1)}), R\} | \hat{v}_{ij} > \hat{v}^{(1)}\right]\Bigg) \end{aligned} \tag{24}$$

whenever $\hat{v}_{ij} \geq Y$. In this equation, $\hat{v}^{(1)} \sim (H^c)^{N-1}$ represents the highest expected value of the other bidders in an auction.

To ease notation, let $O(\hat{v}_{ij}) = H^c(\hat{v}_{ij})^{N-1}$. Then

$$O(\hat{v}_{ij})\mathbb{E}\left[\max\{B(\hat{v}^{(1)}), R\} | \hat{v}_{ij} > \hat{v}^{(1)}\right] = RO(R) + \int_Y^{\hat{v}_{ij}} B(s)O'(s)\,\mathrm{d}s.$$

Substituting into (24) yields

$$V(\hat{v}_{ij})\left(1 - \frac{\lambda_a}{\lambda_a + \rho}\right) = O(\hat{v}_{ij})B(\hat{v}_{ij}) - RO(R) - \int_Y^{\hat{v}_{ij}} B(s)O'(s)\,\mathrm{d}s$$

for $\hat{v}_{ij} \geq Y$. Then substituting into (23)

$$\begin{aligned} B(\hat{v}_{ij}) = \frac{\pi_{\mathbb{J}}}{\lambda_f + \rho}\hat{v}_{ij} \\ - \frac{\rho}{\lambda_f + \rho}\frac{\lambda_a}{\rho}\left[O(\hat{v}_{ij})B(\hat{v}_{ij}) - RO(R) - \int_Y^{\hat{v}_{ij}} B(s)O'(s)\,\mathrm{d}s\right]. \end{aligned}$$

Differentiating with respect to $\hat{v}_{ij}$, I solve explicitly for $B'(\hat{v}_{ij})$.

For $\hat{v}_{ij} \geq Y$,

$$B'(\hat{v}_{ij}) = \frac{\pi_{\mathbb{J}}}{\rho + \lambda_f + \lambda_a H^c(\hat{v}_{ij})^{N-1}}.$$

The marginal bidder obtains zero continuation surplus and therefore

$$B(Y) = R = \frac{\pi_{\mathbb{J}}}{\lambda_f + \rho} Y.$$

Integrating gives

$$B(\hat{v}_{ij}) = R + \pi_{\mathbb{J}} \int_Y^{\hat{v}_{ij}} \frac{1}{\rho + \lambda_f + \lambda_a H^c(s)^{N-1}}\, ds.$$

Bidders with values below Y do not win on the equilibrium path. Their continuation value is therefore zero. In evaluating a deviation to a lower reserve price, their willingness to pay is

$$B(\hat{v}_{ij}) = \frac{\pi_{\mathbb{J}}}{\lambda_f + \rho} \hat{v}_{ij}, \qquad \hat{v}_{ij} < Y.$$

Next, we derive the steady-state distributions H and H^c. Auctions are held at rate A, but an auction results in an ad display only if at least one bidder has a value above the cutoff Y. Thus, the rate at which firms enter a consumer's consideration set is

$$A\left[1 - H^c(Y)^N\right].$$

Matching the total inflow into the consideration set with the total rate of forgetting gives

$$\lambda_f M = A\left[1 - H^c(Y)^N\right].$$

Therefore,

$$M = \frac{A\left[1 - H^c(Y)^N\right]}{\lambda_f}. \tag{25}$$

Conditional on an auction clearing the reserve, the entering firm is the bidder with the highest expected value. The density of the maximum of N independent draws from H^c is

$$NH^c(s)^{N-1} h^c(s).$$

Conditional on the maximum exceeding Y, its density is therefore

$$\frac{NH^c(s)^{N-1} h^c(s)}{1 - H^c(Y)^N}, \qquad s \geq Y.$$

Because firms are forgotten uniformly from the consideration set, the distribution of firms leaving the consideration set is H. Matching the distributions of entering and exiting firms gives

$$h(s) = \frac{NH^c(s)^{N-1}h^c(s)}{1 - H^c(Y)^N}, \qquad s \geq Y. \tag{26}$$

No firm with a value below Y enters a consideration set, so $H(Y) = 0$. Integrating (26) from Y to s gives

$$H(s) = \frac{H^c(s)^N - H^c(Y)^N}{1 - H^c(Y)^N}, \qquad s \geq Y. \tag{27}$$

Recall the accounting identity

$$MH(s) + (J - M)H^c(s) = JG(s). \tag{28}$$

Using (28) to substitute for $H(s)$ in (27), we obtain

$$H^c(s)^N - H^c(Y)^N = \left[\frac{J}{M}G(s) - \frac{J-M}{M}H^c(s)\right]\left[1 - H^c(Y)^N\right], \qquad s \geq Y. \tag{29}$$

Evaluating (28) at $s = Y$ and using $H(Y) = 0$ gives

$$H^c(Y) = \frac{J}{J-M}G(Y). \tag{30}$$

Substituting (30) into (29), the distribution H^c is characterized implicitly by

$$\begin{aligned} H^c(s)^N - \left(\frac{J}{J-M}\right)^N G(Y)^N = &\left[\frac{J}{M}G(s) - \frac{J-M}{M}H^c(s)\right] \\ &\times \left[1 - \left(\frac{J}{J-M}\right)^N G(Y)^N\right], \qquad s \geq Y. \end{aligned} \tag{31}$$

It will also be useful to derive $h^c(Y)$. Evaluating (26) at Y gives

$$NH^c(Y)^{N-1}h^c(Y) = h(Y)\left[1 - H^c(Y)^N\right].$$

Differentiating (28) yields

$$Mh(Y) + (J - M)h^c(Y) = Jg(Y),$$

and hence

$$h(Y) = \frac{J}{M}g(Y) - \frac{J-M}{M}h^c(Y).$$

Substituting into the preceding equation and rearranging gives

$$h^c(Y) = \frac{\frac{J}{M}g(Y)\left[1 - H^c(Y)^N\right]}{NH^c(Y)^{N-1} + \frac{J-M}{M}\left[1 - H^c(Y)^N\right]}. \tag{32}$$

It remains to verify the optimality of the reserve price. Fix the symmetric equilibrium cutoff Y and the associated equilibrium objects H^c and B. Consider a unilateral deviation by one platform to a reserve price that induces cutoff $\hat{Y}$. The deviating platform takes H^c and the equilibrium continuation values of bidders as given.

Let

$$q(y) = N[1 - H^c(y)]H^c(y)^{N-1}$$

denote the probability that exactly one bidder exceeds y, and let

$$g_2(y) = N(N-1)H^c(y)^{N-2}[1 - H^c(y)]h^c(y)$$

denote the density of the second-highest value.

If $\hat{Y} < Y$, the marginal bidder has zero equilibrium continuation value, so the reserve inducing cutoff $\hat{Y}$ is

$$\widehat{R}(\hat{Y}) = \frac{\pi_{\mathbb{J}}}{\lambda_f + \rho}\hat{Y}.$$

Expected profit per auction opportunity is therefore

$$\begin{aligned}\Pi_-(\hat{Y}) = \int_{\hat{Y}}^{\infty} & B(s)g_2(s)\,ds \\ & + \frac{\pi_{\mathbb{J}}}{\lambda_f + \rho}\hat{Y}q(\hat{Y}).\end{aligned}$$

Differentiating and using

$$B(\hat{Y}) = \frac{\pi_{\mathbb{J}}}{\lambda_f + \rho}\hat{Y} \qquad \text{for } \hat{Y} < Y$$

gives

$$\Pi_-'(\hat{Y}) = \frac{\pi_{\mathbb{J}}}{\lambda_f + \rho}NH^c(\hat{Y})^{N-1}\left[1 - H^c(\hat{Y}) - \hat{Y}h^c(\hat{Y})\right].$$

If $\hat{Y} > Y$, the reserve inducing cutoff $\hat{Y}$ is $B(\hat{Y})$. Expected profit per auction opportunity is

$$\begin{aligned}\Pi_+(\hat{Y}) = &\int_{\hat{Y}}^{\infty} B(s) g_2(s)\, ds \\ &+ B(\hat{Y}) q(\hat{Y}).\end{aligned}$$

Differentiating yields

$$\Pi_+'(\hat{Y}) = N H^c(\hat{Y})^{N-1} \left\{ B'(\hat{Y})[1 - H^c(\hat{Y})] - B(\hat{Y}) h^c(\hat{Y}) \right\}.$$

By hypothesis, the virtual value

$$\psi^c(v) = v - \frac{1 - H^c(v)}{h^c(v)}$$

is strictly increasing, and Y solves $\psi^c(Y) = 0$. It follows that

$$\Pi_-'(\hat{Y}) > 0 \qquad \text{for all } \hat{Y} < Y.$$

For $v > Y$, the slope of the bid function,

$$B'(v) = \frac{\pi_{\mathbb{J}}}{\rho + \lambda_f + \lambda_a H^c(v)^{N-1}},$$

is weakly decreasing. Moreover,

$$B(Y) = \frac{\pi_{\mathbb{J}}}{\lambda_f + \rho} Y \geq Y B'(v).$$

Therefore

$$\begin{aligned} B(v) &= B(Y) + \int_Y^v B'(s)\, ds \\ &\geq Y B'(v) + (v - Y) B'(v) = v B'(v). \end{aligned}$$

Thus

$$\frac{B'(v)}{B(v)} \leq \frac{1}{v}.$$

Because $\psi^c(v) > 0$ for $v > Y$,

$$\frac{1}{v} < \frac{h^c(v)}{1 - H^c(v)}.$$

Consequently,

$$\Pi_+'(v) < 0 \qquad \text{for all } v > Y.$$

Hence the candidate cutoff Y globally maximizes the platform's expected profit per auction opportunity.

From here, given the average ad price $\pi_{\mathbb{K}}$, platforms' investment rates are determined as in the baseline model. The condition that $\epsilon - 1 < 1/\varphi$ is used in this step to ensure that all platforms follow the same investment strategy using Lemma 2.

□

Proposition 5. *Suppose that G has compact support $[0, \bar{v}]$ and admits a strictly positive continuous density g. Suppose also that*

$$\frac{A}{\lambda_f} < J.$$

Then the system of steady-state equilibrium conditions in Theorem **??** *admits a solution. Moreover, any regular solution to this system induces a steady-state equilibrium.*

Proof. Fix a candidate reserve cutoff $Y \in [0, \bar{v}]$ and let

$$p(Y) \equiv H^c(Y).$$

Because firms with expected value below Y never enter consideration sets,

$$H(Y) = 0.$$

The accounting identity and the steady-state measure of firms in consideration sets therefore imply

$$p(Y) = \frac{J}{J - M(Y)} G(Y), \tag{33}$$

where

$$M(Y) = \frac{A[1 - p(Y)^N]}{\lambda_f}. \tag{34}$$

Combining (33) and (34), $p(Y)$ must solve

$$p\left[J - \frac{A}{\lambda_f}(1 - p^N)\right] = JG(Y). \tag{35}$$

For each $Y \in [0, \bar{v}]$, equation (35) has a unique solution $p(Y) \in [0, 1]$. Indeed, the left-hand side is strictly increasing in p, since

$$\frac{\partial}{\partial p}\left\{p\left[J - \frac{A}{\lambda_f}(1 - p^N)\right]\right\} = J - \frac{A}{\lambda_f} + \frac{A}{\lambda_f}(N+1)p^N > 0.$$

Moreover, after subtracting $JG(Y)$, the resulting expression is weakly negative at $p = 0$ and weakly positive at $p = 1$. Existence and uniqueness therefore follow from the intermediate value theorem. Because the derivative above is strictly positive, the implicit function theorem implies that $p(Y)$ is continuous. Consequently, $M(Y)$ is continuous as well.

For $s \leq Y$, the accounting identity gives

$$H^c(s) = \frac{J}{J - M(Y)} G(s),$$

and hence

$$h^c(Y) = \frac{J}{J - M(Y)} g(Y).$$

The reserve-price first-order condition is

$$Y = \frac{1 - p(Y)}{h^c(Y)}. \tag{36}$$

Define

$$R(Y) \equiv Y - \frac{1 - p(Y)}{h^c(Y)} = Y - \frac{[1 - p(Y)][J - M(Y)]}{Jg(Y)}.$$

Because p, M, and g are continuous and g is strictly positive, R is continuous on $[0, \bar{v}]$.

At $Y = 0$, equation (35) implies $p(0) = 0$, and therefore

$$M(0) = \frac{A}{\lambda_f}.$$

It follows that

$$R(0) = -\frac{J - A/\lambda_f}{Jg(0)} < 0.$$

At the upper endpoint, $G(\bar{v}) = 1$. Equation (35) then implies $p(\bar{v}) = 1$, so

$$R(\bar{v}) = \bar{v} > 0.$$

The intermediate value theorem therefore yields some $Y^* \in (0, \bar{v})$ satisfying (36).

It remains to construct the distributions associated with Y^*. Write $p^* = p(Y^*)$ and $M^* = M(Y^*)$. For $s \leq Y^*$, define

$$H^c(s) = \frac{J}{J - M^*} G(s).$$

For $s \geq Y^*$, combine the stationarity condition

$$H^c(s)^N - p^{*N} = H(s)(1 - p^{*N})$$

with the accounting identity

$$M^* H(s) + (J - M^*) H^c(s) = JG(s)$$

to obtain

$$H^c(s)^N - p^{*N} = \left[\frac{JG(s) - (J - M^*) H^c(s)}{M^*} \right] (1 - p^{*N}). \tag{37}$$

For each $s \in [Y^*, \bar{v}]$, the residual associated with (37) is strictly increasing in $H^c(s)$, since its derivative is

$$NH^c(s)^{N-1} + \frac{J - M^*}{M^*}(1 - p^{*N}) > 0.$$

At $s = Y^*$, equation (37) is satisfied by $H^c(Y^*) = p^*$, while at $s = \bar{v}$ it is satisfied by $H^c(\bar{v}) = 1$. Thus (37) uniquely defines $H^c(s) \in [p^*, 1]$.

Implicit differentiation yields

$$h^c(s) = \frac{\frac{J}{M^*} g(s)(1 - p^{*N})}{NH^c(s)^{N-1} + \frac{J - M^*}{M^*}(1 - p^{*N})} > 0.$$

Hence H^c is a valid continuously differentiable distribution function. The distribution H is then recovered from the accounting identity. The bid function, expected ad price, investment rate, and steady-state quality are constructed exactly as in Theorem **??**. Thus the system of steady-state equilibrium conditions admits a solution.

Finally, Theorem **??** shows that any regular solution to this system satisfies firms' bidding optimality, platforms' reserve-price and investment optimality, stationarity, and all market-clearing conditions. Therefore any regular solution induces a steady-state equilibrium. □

I conjecture that if G is supported on a compact interval, admits a continuously differentiable density bounded away from zero, and has strictly increasing virtual values, then for J sufficiently large relative to A, the distribution H^c induced by any solution to the steady-state equilibrium system has a strictly increasing virtual value function. Intuitively, as the mass of firms becomes large relative to the rate at which ads are displayed, the distribution of firms outside consumers' consideration sets converges to the population distribution G, thereby inheriting its regularity.

The following proposition shows that ad revenue is maximal when data is uninformative, across all steady-state equilibria.

Proposition 6. *Let $\{G_n\}$ be a sequence of continuous cdfs converging pointwise to the Heaviside function centered at μ_F:*

$$\lim_{n\to\infty} G_n(\hat{v}) = \mathbb{1}_{\{\hat{v}\geq\mu_F\}} \qquad \text{for every } \hat{v} \in [0,\infty).$$

For each n, let $\pi^R_{\mathbb{K}}(G_n)$ denote steady-state ad revenue in the reserve-price extension, where platforms optimally choose their reserve prices. Then

$$\lim_{n\to\infty} \pi^R_{\mathbb{K}}(G_n) = \sup_G \pi^R_{\mathbb{K}}(G) = \frac{I}{\sigma M(\rho+\lambda_f)},$$

where the supremum is over all continuous cdfs G supported on $[0,\infty)$.

Moreover, along any such sequence of equilibria, the probability that an auction fails because the reserve price is not met converges to zero.

Proof. Fix an arbitrary continuous cdf G and consider any steady-state equilibrium of the reserve-price extension. Let P denote the payment made following a successful auction and let $\hat{v}^{(1)}$ denote the winning advertiser's expected value. Because the auction is individually rational, the winner's payment cannot exceed its bid. Moreover, the bid cannot exceed the value obtained if the advertiser were never displaced by another advertisement. Therefore,

$$P \leq B(\hat{v}^{(1)}) \leq \frac{\pi_{\mathbb{J}}}{\rho+\lambda_f}\hat{v}^{(1)}.$$

Conditional on an advertisement being displayed, the winning advertiser's value is distributed according to H. Hence

$$\pi^R_{\mathbb{K}}(G) = \mathbb{E}[P] \leq \frac{\pi_{\mathbb{J}}}{\rho+\lambda_f}\mu_H.$$

Using

$$\pi_{\mathbb{J}} = \frac{I}{\sigma M\mu_H},$$

we obtain

$$\pi^R_{\mathbb{K}}(G) \leq \frac{I}{\sigma M(\rho+\lambda_f)}.$$

Thus,

$$\sup_G \pi^R_{\mathbb{K}}(G) \leq \frac{I}{\sigma M(\rho+\lambda_f)}.$$

Now consider the sequence $\{G_n\}$. Because G_n converges to a point mass at μ_F, the distributions of advertisers' expected values inside and outside consumers' consideration sets also converge to a point mass at μ_F. Consequently, the distribution of bids converges to a point mass at

$$\bar{b} = \frac{\pi_{\mathbb{J}}\mu_F}{\rho + \lambda_f} = \frac{I}{\sigma M(\rho + \lambda_f)}.$$

Let Y_n denote the equilibrium reserve and let

$$\alpha_n = H_n^c(Y_n)^N$$

be the probability that the reserve is not met. Suppose, toward a contradiction, that along some subsequence

$$\alpha_n \to \alpha > 0.$$

Since the bid distribution converges to a point mass at $\bar{b}$, any reserve that is met with nonvanishing probability must itself converge to $\bar{b}$. The payment conditional on a successful auction therefore converges to $\bar{b}$, while an advertisement is displayed with probability converging to $1 - \alpha$.

Consider instead a deviation to a nonbinding reserve. Under this deviation, an advertisement is displayed with probability converging to one, and the payment converges to $\bar{b}$. Because an auction opportunity generates the same nuisance cost whether or not the reserve is met, failing to display an advertisement provides no offsetting benefit. The deviation therefore raises limiting revenue from $(1 - \alpha)\bar{b}$ to $\bar{b}$, contradicting the optimality of Y_n.

It follows that

$$\alpha_n \to 0.$$

Thus the equilibrium reserve is asymptotically nonbinding. Since bids converge to $\bar{b}$ and the probability of a successful auction converges to one,

$$\lim_{n\to\infty} \pi_{\mathbb{K}}^R(G_n) = \bar{b} = \frac{I}{\sigma M(\rho + \lambda_f)}.$$

Combining this equality with the upper bound proves

$$\lim_{n\to\infty} \pi_{\mathbb{K}}^R(G_n) = \sup_G \pi_{\mathbb{K}}^R(G).$$

□

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